EVQ-Cosh: Variational Frequency Allocation for Rotary Position Embedding

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Abstract

Rotary position embedding (RoPE) is not only a positional operator but also a finite spectral budget: each head has only $d_{\text{head}}/2$ rotary frequency channels, yet the standard geometric schedule spends this budget by inherited convention. We study training-time frequency-channel allocation as a RoPE design axis complementary to operator design and inference-time range scaling. Formulating allocation as a variational inverse problem over log-frequency densities, we derive EVQ-Cosh, a closed-form one-parameter family in which geometric RoPE is recovered as the $\tau \to 0$ limit of the pure-tether sub-family under a broadband collision surrogate. EVQ-Cosh changes only the inverse-frequency initialization and adds zero learned parameters. A semi-analytic operating-point rule $\tau^* = d_{\text{head}}/\sqrt{L}$ selects a non-geometric allocation; its d_{head} factor and $L^{-1/2}$ exponent are predicted under a small-τ softmax-transport model with χ^2 channel-load stiffness, and the remaining $\mathcal{O}(1)$ normalization is fixed by convention inside an empirically flat PPL basin. Three primary stress tests anchor our evidence: in PE-dominant DAPE-style 128→8K extrapolation, EVQ improves over geometric RoPE and a learned positional operator; under matched YaRN scale on a 454M passkey-mix model, EVQ+YaRN reaches 100% passkey retrieval at 8K while Geo+YaRN remains at 61%; and on a 432M MLA model with 16 frequency channels, EVQ alone reduces 2× extrapolation PPL by 31.1% at +1.1% in-distribution cost, surpassing matched-scale (s=4) Geo+YaRN. These results support training-time allocation *shape*, not only frequency *range*, as a first-order RoPE design axis.

1 Introduction

Rotary position embedding (RoPE; Su et al., 2024) is often discussed as a positional operator, but in implementation it is also a *finite spectral allocation*: each attention head receives only $d_{\text{head}}/2$ rotary frequency channels, and the default geometric (equal-ratio) schedule is one inherited convention for spending this budget. This convention propagates unchanged from LLaMA-3 [Grattafiori et al., 2024] and Qwen2.5 [Yang et al., 2024a] to DeepSeek-V3 [DeepSeek-AI, 2024b] and video DiTs [Wei et al., 2025], even as model scales, context lengths, and architectures vary. Once RoPE is viewed as a finite spectral budget, a basic question appears: before training begins, how should these channels be allocated?

Most long-context work answers a different question. Base-frequency tuning changes the span of the spectrum; positional interpolation [Chen et al., 2023], NTK-aware scaling [Rozière et al., 2023], YaRN [Peng et al., 2024], and LongRoPE-style methods [Ding et al., 2024, Shang et al., 2025] rescale positions at or after inference. These *range-scaling* methods

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stretch or reuse a frequency table whose allocation was fixed at initialization. Adjacent works observe that the allocation matters empirically—Barbero et al. [2025] find high- and low-frequency specialization after training, Wei et al. [2025] propose a modality-specific allocation for video, Wang et al. [2024] preserve critical frequencies during interpolation—but none gives a closed-form *training-time* allocation for standard RoPE derived from an explicit variational surrogate.

We treat training-time frequency-channel allocation as a third RoPE design axis, complementary to operator and range. Parameterizing allocations by densities $\rho(\phi)$ over log-frequency, we introduce EVQ-Cosh: a closed-form one-parameter allocation family with zero learned parameters that changes only the RoPE inverse-frequency initialization. Geometric RoPE is recovered as the $\tau \to 0$ limit of the pure-tether sub-family (not a generic optimum). A small-τ softmax-transport calculation motivates the deployable operating rule $\tau = d_{\text{head}}/\sqrt{L}$, used as a robust default inside an empirically flat basin rather than a claim of global optimum.

Empirical scope. We report three *primary* stress tests (Primary I–III in §4, 3-seed where indicated): a matched-scale substrate test composing EVQ with YaRN on a 454M transformer (3-seed); a PE-dominant 128→8K extrapolation test against DAPE [Zheng et al., 2024a] (1–3 seed; the learned-PE row is 3-seed, EVQ/Geo/DAPE rows are seed 42); and a scarce-channel test on MLA [DeepSeek-AI, 2024a,b] with 16 RoPE channels at 432M (3-seed). We additionally report *robustness* experiments on the same MHA setting, and *supporting/exploratory* single- or two-seed evidence (video DiT, LoRA into LLaMA-3-8B, progressive training) that extends scope rather than anchoring primary claims; tier assignment appears in Table 3.

Contributions.

- **Finite-spectral-budget view.** We identify RoPE training-time frequency-channel allocation as a third design axis, complementary to operator design and inference-time range scaling; geometric RoPE is an inherited allocation convention, not a consequence of the RoPE operator.
- **Closed-form allocation family.** We introduce EVQ-Cosh: a closed-form one-parameter inverse-CDF allocation family with zero learned parameters, in which geometric RoPE is recovered as the $\tau \to 0$ limit of the pure-tether sub-family. The method modifies only the inverse-frequency initialization.
- **Deployable operating-point rule.** A small-τ softmax-transport argument gives the structural scaling $\tau^* \propto d_{\text{head}}/\sqrt{L}$ with convention-dependent prefactor; $\tau = d_{\text{head}}/\sqrt{L}$ is used as a robust default inside an empirically flat basin rather than a claimed global optimum.
- **Three mechanism stress tests.** Matched-scale EVQ+YaRN tests substrate–range complementarity; PE-dominant extrapolation against DAPE tests whether allocation shape alone closes extrapolation gaps; compressed-RoPE MLA is the scarce-channel regime where allocation quality should matter most.

2 Related Work

Positional encodings can be separated along three axes: the *operator* that injects positional information, the *inference-time* mechanism that extends its usable range, and the *frequency-usage* analysis of what channels actually do after training. We position EVQ-Cosh in each and summarize the landscape in Table 1.

Axis A: Positional-encoding operators. Transformers rely on positional structure to recover order [Vaswani et al., 2017]; the design space spans absolute encodings, relative biases [Shaw et al., 2018, Raffel et al., 2020], additive linear biases (ALiBi; Press et al., 2022), kernelized biases (KERPLE; Chi et al., 2022), rotary position embedding (RoPE; Su et al., 2024), magnitude-decay rotary (XPOS; Sun et al., 2023), and learned positional functions (FIRE, Li et al., 2024; DAPE, Zheng et al., 2024a). Hyperbolic-RoPE [Dai et al.,

2025] (distinct from the VLM-targeted HoPE [Li et al., 2025]) replaces the SO(2) rotation with hyperbolic Lorentz boosts, putting cosh/sinh *inside* the operator. EVQ-Cosh is not a new operator: it preserves the RoPE rotation and uses cosh only as the closed-form solution of a variational ODE on log-frequency density—modifying how the fixed number of frequency channels is distributed along the spectrum. In the PE-dominant regime (§4.3) this zero-learned-parameter allocation change closes more of the extrapolation gap than a learned positional-operator baseline, isolating the frequency-structure effect from operator capacity.

Axis B: Inference-time and post-training context extension. NTK-aware scaling [Rozière et al., 2023], positional interpolation [Chen et al., 2023], and YaRN [Peng et al., 2024] rescale RoPE positions on a frozen checkpoint; SelfExtend [Jin et al., 2024] uses grouped attention without fine-tuning; LongRoPE [Ding et al., 2024] and PoSE [Zhu et al., 2024] use progressive or skip-wise strategies; LongRoPE2 [Shang et al., 2025] searches per-channel scaling factors by evolution on a frozen checkpoint and combines this with mixed-context-window adaptation; CLEX [Chen et al., 2024] trains a continuous extrapolation module. All of these preserve the pretrained checkpoint and modify behavior at or after inference. EVQ-Cosh is orthogonal: it changes the *training-time* frequency substrate itself, and in our experiments (§4.2) composes with YaRN under matched YaRN scale. *Relation to LongRoPE2.* LongRoPE2 is the closest post-hoc per-channel rescaling method to EVQ, but the two target different stages: LongRoPE2 searches scaling factors for a pretrained checkpoint, whereas EVQ specifies the training-time frequency allocation before optimization begins. The methods are therefore conceptually composable; we leave empirical EVQ+LongRoPE2 composition to future work.

Axis C: Frequency-usage and allocation analysis. Barbero et al. [2025] show empirically that high-frequency channels specialize in positional patterns while low-frequency channels carry semantic content, an asymmetry that geometric allocation does not model. Wang et al. [2024] (Resonance RoPE) identify critical frequencies that should not be perturbed during interpolation, implicitly acknowledging channel inequality but framed around interpolation rather than allocation. Wei et al. [2025] propose sequence-length-dependent allocation tailored for video generation; Zhao et al. [2025] correct frequencies at inference time for video diffusion; Qiu et al. [2024] addresses temporal coherence via noise rescheduling (adjacent, not a positional-encoding method). Concurrent work covers adjacent facets: MHRoPE [Huang et al., 2026] and HoPE [Li et al., 2025] tune axis-split allocations for multimodal RoPE in vision–language models; FoPE [Hua et al., 2025] models each channel as a Fourier series; CARoPE [Veisi et al., 2025] learns dynamic per-head frequencies; Clipped RoPE [Li et al., 2026] soft-clips low frequencies post-hoc. None derives a closed-form training-time spectral density for standard RoPE from a variational principle.

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024, Hsieh et al., 2024]. Prior work shows that long contexts can fail even when relevant information is nominally present [Liu et al., 2024], and that positional encoding modifications can mitigate such failures [Zhang et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

Positioning. EVQ-Cosh occupies a previously unfilled cell (Table 1): a training-time, closed-form, zero-learned-parameter frequency-allocation substrate. It does not compete with range-extension methods by offering another rescaling rule, nor with learned PE methods by adding operator capacity; it changes the pretraining substrate—the frequency table that all subsequent RoPE-based range-scaling and operator-adaptation methods inherit. Per-channel learned/searched methods [Shang et al., 2025, Veisi et al., 2025, Hua et al., 2025] target a distinct stage (post-hoc adaptation of a frozen checkpoint); empirical head-to-head benchmarking is left to future work.

Table 1: Compact comparison of RoPE-family context methods. EVQ-Cosh is the only row whose main axis is closed-form variational training-time frequency allocation.

Method	Train alloc.	Infer scale	Post-hoc/ search	Learned	Closed form	Main axis
Geometric RoPE [Su et al., 2024]	fixed	—	—	—	✓	default operator
NTK / PI [Rozière et al., 2023, Chen et al., 2023]	—	✓	post-hoc	—	✓	global range
YaRN [Peng et al., 2024]	—	✓	post-hoc	—	✓	piecewise range
LongRoPE [Ding et al., 2024]	—	✓	search	—	—	range schedule
LongRoPE2 [Shang et al., 2025]	—	✓	search	—	—	per-channel range
DAPE [Zheng et al., 2024a]	—	—	training	✓	—	adaptive operator
FIRE [Li et al., 2024]	—	—	training	✓	—	learned function
VideoRoPE [Wei et al., 2025]	heuristic	✓	design	—	✓	modality-specific
Resonance RoPE [Wang et al., 2024]	—	✓	design	—	✓	critical frequencies
MHRoPE/HoPE [Huang et al., 2026, Li et al., 2025]	heuristic	—	design	—	✓	VLM axis-split
CARoPE [Veisi et al., 2025]	—	—	training	✓	—	learned dynamic frequencies
EVQ-Cosh (ours)	✓	—	—	—	✓	**train-time allocation**

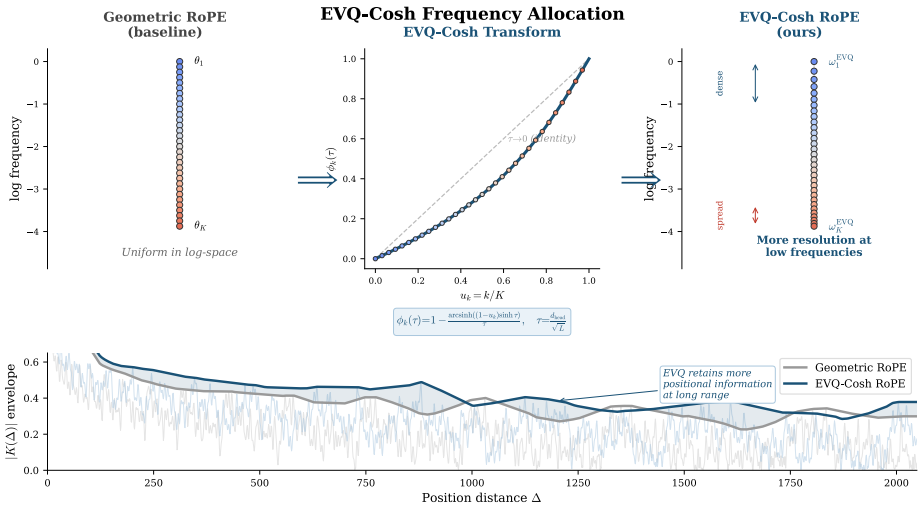

Figure 1: **EVQ-Cosh frequency allocation.** Left: geometric RoPE places channels uniformly in log-frequency. Center: the EVQ-Cosh warp $\phi_k(\tau)$ bends below the identity (dashed); $\tau \to 0$ recovers geometric RoPE. Right: the allocation *widens spacing* among low-frequency channels and *extends* the spectrum toward longer wavelengths, reducing long-range collisions.

3 Theory and Method

Before any derivation we state the epistemic status of each component (Table 2), so that the variational core is not conflated with the practical recipe.

The surrogate is functional, not pointwise (Appendix A.5: 24–92% collision-score reduction across 12 configs); downstream PPL/passkey then measures the trained-model consequence, not $\mathcal{C}_{\text{app}}$ itself. The pure-tether branch of the stationary ODE (Theorem 1) is a design choice that keeps the allocation invertible in closed form; it is not a claim that the Fisher-driven forcing term never matters. The rule $\tau^* = d_{\text{head}}/\sqrt{L}$ (Proposition 1) is an operating-point selector whose d_{head} factor and $L^{-1/2}$ exponent are derived; its $\mathcal{O}(1)$ scale is by convention, moving the allocation away from the $\tau = 0$ boundary into a wide non-geometric basin.

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency, $\omega(\phi) = b^{-\phi}$. Let d_{rot} denote the number of dimensions to which RoPE is applied and $K = d_{\text{rot}}/2$ the number of rotary frequency pairs. For standard MHA, $d_{\text{rot}} = d_{\text{head}}$; for compressed-RoPE architectures such as MLA (§4.4), d_{rot} may be smaller than the full attention head dimension d_{head}, in which case the inverse-CDF grid uses $K = d_{\text{rot}}/2$ while the operating-point scale uses the effective attention-head dimension d_{eff} specified for that architecture (we take $d_{\text{eff}} = d_{\text{head}}$ for MLA;

Table 2: Epistemic map of the EVQ-Cosh derivation. The variational core (rows 1–2) is exact conditional on the broadband surrogate; the practical recipe (rows 3–4) is a deliberate engineering choice whose residual uncertainty is bounded empirically (§3.7).

Component	Status	Role
Broadband surrogate $\mathcal{C}_{\text{app}}$	Functional approximation of the *collision objective* (not pointwise approximation of the kernel K)	Yields a tractable variational problem; validated in Appendix A.5
ODE and closed-form family	Exact conditional on $\mathcal{C}_{\text{app}}$	Theoretical core; geometric RoPE is the $\tau\to 0$ limit of the pure-tether sub-family
Pure-tether branch	Practical branch choice of the stationary ODE	Gives a closed-form invertible quantile warp; forcing term suppressed at typical bases
τ^* scaling rule	Semi-analytic operating-point selector (not a global minimizer)	Moves allocation away from the degenerate $\tau\to 0$ edge into a wide non-geometric basin

see §4.4 and Appendix C.1). We use a positive probability density $\rho \in C^2([0,1])$ with $\int_0^1 \rho(\phi)\, d\phi = 1$ to parameterize a family of discrete RoPE allocations via inverse-CDF quantization: $\phi_k = F_\rho^{-1}(u_k)$ with $F_\rho(\phi) = \int_0^\phi \rho(s)\, ds$ and grid u_k (Figure 1). Geometric RoPE corresponds to $\rho \equiv 1$.

3.2 A surrogate collision functional

Under a distance prior $D(\Delta)$, the exact phase-collision kernel is $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$; it is oscillatory and is not pointwise approximated by a smooth two-term model. We therefore introduce the following surrogate collision quadratic form as the object of our variational analysis:

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\, d\phi + \frac{\beta}{2}\iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi,\psi)\, d\phi\, d\psi, \tag{1}$$

formally corresponding to the distributional kernel $K_{\text{app}}(\phi,\psi) := \alpha\,\delta(\phi-\psi) + \beta \min(\phi,\psi)$. The diagonal term penalizes channel-load concentration; $\min(\phi,\psi)$ is the Green kernel of $-\partial_\phi^2$ on $[0,1]$ and encodes cumulative low-frequency overlap. We validate $\mathcal{C}_{\text{app}}$ *functionally*, not pointwise: allocations derived from $\mathcal{C}_{\text{app}}$ reduce the exact-kernel collision score by 24–92% across 12 configurations, and a collision-only oracle search reaches comparable reductions (Appendices A.5, A.13).

3.3 Constrained stationary allocation

Minimizing $\mathcal{C}_{\text{app}}[\rho]$ in (1) subject to $\int_0^1 \rho(\phi)\, d\phi = 1$ with Lagrange multiplier ν, the first variation gives $\alpha\rho(\phi) + \beta\, g(\phi) + \nu = 0$ where $g(\phi) = \int_0^1 \rho(\psi) \min(\phi,\psi)\, d\psi$. Since $g''(\phi) = -\rho(\phi)$, every interior stationary density satisfies the homogeneous equation $\rho''(\phi) - \tau^2\rho(\phi) = 0$ with $\tau = \sqrt{\beta/\alpha}$. The pre-differentiated stationarity and mass normalization give the derivative boundary conditions $\rho'(0) = -\tau^2$ and $\rho'(1) = 0$ (Appendix A).

Theorem 1 (Exact stationary allocation under the broadband surrogate)**.** *Under* (1) *with* $\alpha{>}0, \beta{\geq}0$ *(so* $\mathcal{C}_{\text{app}}$ *is convex; Appendix A) and constraints* $\rho \in C^2([0,1])$, $\rho{>}0$, $\int_0^1 \rho{=}1$, *the unique constrained minimizer is*

$$\rho_\tau(\phi) \;=\; \frac{\tau\,\cosh(\tau(1-\phi))}{\sinh\tau}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{2}$$

strictly positive on $[0,1]$ *for* $\beta{>}0$*; at* $\beta{=}0$ *the minimizer is the* $\tau\to 0$ *limit* $\rho_0 \equiv 1$ *(uniform/geometric).*

Fisher-forcing extension (Appendix). Adding a Fisher utility $-\mu\int \rho\, b^{-2\phi}\, d\phi$ to (1) changes the stationary equation to $\rho'' - \tau^2\rho = \gamma\, b^{-2\phi}$, whose non-resonant solution includes a particular term $\gamma\, b^{-2\phi}/((2\log b)^2 - \tau^2)$. EVQ-Cosh corresponds to the homogeneous $\gamma{=}0$ branch; the forced extension is analyzed as a perturbation in Appendix A.15. Our practical method uses the homogeneous branch because it gives a closed-form invertible CDF; the forced branch is a controlled but nonzero residual, not zero pointwise.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,, \qquad u_k = \frac{k}{K}, \tag{3}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2 (Closed-form pure-tether allocation and geometric limit)**.** *For $\tau > 0$, the stationary density in Theorem 1 (the homogeneous, pure-tether branch) has CDF $F_\tau(\phi) = 1 - \sinh(\tau(1{-}\phi))/\sinh\tau$, so inverse-CDF quantization gives $\phi_k(\tau) = 1 - \tau^{-1}\operatorname{arcsinh}((1{-}u_k)\sinh\tau)$. As $\tau \to 0$, $\phi_k(\tau) \to u_k$ and geometric RoPE is recovered* within the pure-tether sub-family*; the Fisher-forced extension (Appendix A.2) does not vanish in this limit.*

3.5 Practical EVQ instantiation

For an effective attention-head dimension d_{eff} and training length L, the complete implementation is

$$\tau = \frac{d_{\text{eff}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{rot}}}{2}. \tag{4}$$

For standard MHA, $d_{\text{eff}}{=}d_{\text{rot}}{=}d_{\text{head}}$; for compressed-RoPE MLA, $K{=}d_{\text{rot}}/2$ counts only the rotary channels while d_{eff} follows the architecture-specific convention (we take $d_{\text{eff}}{=}d_{\text{head}}$; see §4.4 and Appendix C.1). The construction is deliberately minimal: set τ, compute quantiles, evaluate the warp, replace inverse frequencies. No auxiliary loss, no extra parameters, no optimizer dependence. The endpoint grid $u_k{=}k/K$ has the anchoring property $u_0{=}0{\Rightarrow}\phi_0{=}0$ (preserves $\omega_0{=}1$); our implementation uses midpoint quantization $u_k{=}(2k{+}1)/(2K)$, which trades exact anchoring for avoiding endpoint cells. We do not rely on $\omega_0{=}1$ in any trained model, and the endpoint/midpoint choice shifts PPL by $<1\%$ across all $K{\geq}16$ settings we tested, well inside the flat empirical basin.

EVQ-Cosh changes only the inverse-frequency initialization—distinct from learnable PE approaches that rely on in-distribution training dynamics to discover the allocation through extra parameters. The full recipe is Equation (4) plus $\tau{=}d_{\text{eff}}/\sqrt{L_{\text{train}}}$: zero learned parameters, architecture unchanged.

3.6 Waterbed inequality

Under the broadband surrogate, EVQ obeys a waterbed inequality (a corollary of the sharper envelope in Proposition 3, Appendix A.7). Writing P_ρ for the CDF-induced measure and U for the uniform distribution on $[0,1]$, and letting $D_{\text{Burg}}(\rho){:=}{-}\int_0^1 \log\rho(\phi)\, d\phi$ denote the Burg entropy deficit relative to uniform, the Pearson-divergence companion functional $\mathcal{W}(\rho) = \alpha\chi^2(U\|P_\rho) + \beta\int M_\rho^2$ (related to but distinct from $\mathcal{C}_{\text{app}}$: both vanish at $\rho{\equiv}1$ with leading-order $\mathcal{O}(\tau^4)$ growth; they diverge at large τ) satisfies $\mathcal{W}(\rho) \geq \alpha(e^{D_{\text{Burg}}(\rho)} - 1)$ with equality iff $\rho \equiv 1$. Since $\mathcal{W}(\rho_\tau){=}\mathcal{O}(\tau^4)$ and Proposition 1 (with the $L \to L_{\text{eff}}^J$ substitution from Appendix A.14) gives $\tau_*^2{\sim}d_{\text{head}}^2/L_{\text{eff}}^J$, EVQ's in-range Pearson deviation scales as $\mathcal{W}(\rho_{\tau_*}) = \mathcal{O}(d_{\text{head}}^4/(L_{\text{eff}}^J)^2)$. Long-range improvement is thus paid for by a bounded, quartic-in-τ^* in-range cost.

3.7 The τ^* scaling law: epistemic stratification

Shape and scale come from separate layers. The cosh shape is the pre-softmax stationary solution under $\mathcal{C}_{\text{app}}$ (§3.3–§3.5); the scale τ^* is selected *post*-softmax by the

proposition below, with $U(\tau, L)$ the per-channel post-softmax KL gain from an $\mathcal{O}(\tau^2)$ RoPE logit perturbation (formal definition Appendix A.11). The $1/L$ factor is the softmax Jacobian eigenvalue at the diffuse baseline $p_0{=}1/L$—an algebraic property, not a CLT claim.

Proposition 1 (Semi-analytic operating-point scaling under a diffuse baseline)**.** *On the pure-tether family ρ_τ, and assuming a diffuse post-softmax baseline $p_0{=}1/L$, the Pearson-χ^2 channel-load stiffness and post-softmax transport utility admit small-τ expansions*

$$S_{\chi^2}(\tau) \;=\; c_S\,\frac{\tau^4}{d_{\text{head}}} + O(\tau^6), \quad c_S{=}\tfrac{1}{45}, \qquad U(\tau, L) \;=\; \frac{d_{\text{head}}}{L}\big[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)\big],$$

with $Q_1(L, b) = \int_0^1 \eta(\phi)\, q(Lb^{-\phi})\, d\phi$, $\eta(\phi) = (1{-}\phi)^2/2{-}1/6$, and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$ (Appendix A.11). The stationary point of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda\, U(\tau, L)$ therefore satisfies

$$\tau_*^2 \;=\; 45\,\lambda\, Q_1(L, b)\cdot\frac{d_{\text{head}}^2}{L}, \qquad \tau_* \;\propto\; \frac{d_{\text{head}}}{\sqrt{L}}.$$

The d_{head} factor and $L^{-1/2}$ exponent follow from the combined small-τ scalings; the rule balances an $\mathcal{O}(\tau^2)$ utility gain (RoPE logit perturbation transported through a diffuse softmax with $1/L$ Jacobian curvature) against an $\mathcal{O}(\tau^4)$ channel-load stiffness. The remaining $\mathcal{O}(1)$ prefactor (c_S, λ, Q_1) depends on normalization convention and is fixed empirically inside a flat basin, not claimed as a global optimum.

Three epistemic tiers. The small-τ expansion in Proposition 1 *derives the structural* d_{head} factor and $L^{-1/2}$ exponent; at practical τ (e.g., $\tau{=}4$ in the PE-dominant setting) the exact cosh family is used directly, not the Taylor truncation, and numerical optimization yields an exponent within 1% PPL of $L^{-0.500}$. The Pearson χ^2 stiffness $S_{\chi^2}(\rho){=}\mathrm{Var}_{\rho d\phi}(1/\rho)$ is the canonical load-variance penalty under inverse-CDF cell-width $1/\rho(\phi)$ (Appendix A.16); the Lagrange multiplier $\lambda{=}1$ is a unit convention, with an exact-kernel sweep giving $c_{\text{coll}}{=}1.171$ vs. surrogate $\sqrt{45Q_1}{\approx}1.19$ (1.6% relative, CV 0.28%, 9 configs; Table 7). For trained, non-diffuse attention the same expansion replaces $1/L$ by a softmax-Jacobian Rayleigh quotient $1/L_{\text{eff}}^J$ (Appendix A.14); we register direct measurement at $L_{\text{train}}{\geq}16$K as a falsifiable test of the diffuse-baseline approximation.

Basin, not point estimate. With $\lambda{=}1$, the formula falls within $<1\%$ PPL of the best observed sweep points across 27 configurations; pure geometric ($\tau{=}0$) is the natural negative control and falls 10–46% outside (Appendices A.10, A.9). The bare formula stays inside this basin for our primary $b{\geq}500$K regime; at small b and large L the explicit $c_{\text{pred}}(L, b){=}\sqrt{45\,Q_1(L, b)}$ form is recommended. EVQ and YaRN correct orthogonal deficiencies (shape vs. range; Appendix Fig. 3), predicting the additive EVQ+YaRN composition in §4.

4 Experiments

4.1 Setup and evidence tiering

Evaluation scope. This is a positional-encoding mechanism study. PE-diagnostic endpoints (extrapolation PPL, passkey, NIAH/RULER-style retrieval, gold-answer NLL) are our causal targets; downstream accuracy at these scales is confounded with semantic capacity and is reported as a non-regression check rather than a primary claim. EVQ-Cosh modifies only the training-time allocation and adds no learned parameters—it is neither an inference-time rescaler (YaRN/NTK/LongRoPE) nor a learned positional operator (DAPE/FIRE).

Experiments test three questions derived from the variational characterization: (Q1) does EVQ improve over Geo at fixed training length? (Q2) does EVQ compose with inference-time scaling, i.e., does EVQ+YaRN outperform Geo+YaRN at extrapolated lengths under matched YaRN scale? (Q3) does the τ^* rule land in the best observed sweep basin across architectures and scales? Core experiments train from scratch with fixed architecture, modifying only the RoPE inverse-frequency initialization. We use $b{=}500$K, the predicted $\tau^*(L)$ for each training length, and standard GPT-style decoder stacks [Radford et al., 2019] with $d_{\text{head}}{=}64$

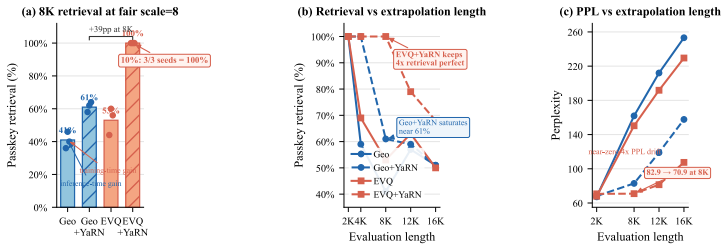

Figure 2: EVQ and YaRN correct orthogonal deficiencies. EVQ expands low-frequency spacing at training time; YaRN rescales positions at inference time. Their combination under matched YaRN scale substantially outperforms Geo+YaRN on both retrieval and PPL.

across 50M–750M. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings. Full hyperparameters are in Appendix D.

Evidence is organized in three tiers (Table 3). *Primary* (directly tied to central claims): EVQ×YaRN composition (§4.2, 3-seed), PE-dominant extrapolation (§4.3, 1–3 seed; only the learned-PE comparison row is 3-seed), and MLA (§4.4, 3-seed). *Robustness* (3-seed): FineWeb-Edu consistency, base sweep, and capability checks (§4.5). *Supporting / exploratory* (1–2 seed): one-seed multi-scale rows, video DiT, LoRA into LLaMA-3-8B, progressive training, and MLA 1B-token (Appendix D). No abstract or introduction claim rests on tier-3 evidence.

Table 3: Evidence tier by setting. Primary claims rest only on 3-seed primary tier; robustness rows are 3-seed but not primary anchors.

Tier	Setting	Seeds	Location
Primary I	EVQ+YaRN, 454M MHA, L_{train}=2048, 4/6/8× extrapolation	3	§4.2
Primary II	PE-dominant 128→8K (DAPE; learned-PE row 3-seed, others seed 42)	1–3	§4.3
Primary III	MLA 432M, 16 RoPE channels, 500M tokens	3	§4.4
Robustness	FineWeb-Edu consistency, base sweep, capability preservation	3	§4.5
Supporting	Multi-scale raw-PPL one-seed rows	1	§4.5, App. D
Supporting	Video DiT 129M / 382M	1–2	§4.5, App. D
Supporting	LoRA into LLaMA-3-8B-Instruct	1	§4.5, App. D
Supporting	Progressive training	1	§4.5, App. D
Supporting	MLA 1B-token (robustness to training saturation)	1	§4.4, App. D

4.2 Primary I: matched-scale substrate test (EVQ vs. YaRN)

Substrate vs. range stress test. If EVQ only changed effective range, applying the same YaRN scale should largely erase its advantage; if it changes the training-time substrate, the same rescaling rule should have higher leverage on EVQ than Geo (Table 4, Fig. 2).

In the 454M 10% passkey-mix setting (3 seeds per config), EVQ+YaRN reaches 100% passkey retrieval at 8K while Geo+YaRN remains at 61%, with the advantage persisting at 12/16K (79%/68% vs. 59%/51%; Table 4); PPL drops from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. EVQ alone improves over Geo without saturating retrieval, so the jump to 100% under YaRN is not a ceiling effect. Table 20 (Appendix) quantifies the interaction: YaRN gives Geo only $-4.5\%/-3.1\%$ at 4/8K but gives EVQ (τ=4) $-32.5\%/-40.7\%$; naive NTK-aware scaling becomes destructive at large τ. This is consistent with substrate–range complementarity: EVQ corrects shape, YaRN corrects range, and their corrections are additive (Figure 2; Appendix Fig. 3). We read the result as EVQ increasing YaRN's leverage under matched scale; we do not claim dominance over every tuned-scale YaRN baseline.

4.3 Primary II: PE-dominant allocation stress test

Allocation-shape stress test. In a DAPE-style 128→8K protocol [Zheng et al., 2024a] where model capacity is deliberately limited, we ask whether allocation shape alone can close a PE-dominant extrapolation gap that a learned positional operator cannot. Table 5 (L_{train}=128, d_{head}=64): EVQ reaches 333.7 PPL (vs. Geo 513.7, DAPE 455.3) with no learned parameters; the closed-form reallocation (zero learned parameters) closes more of

Table 4: Training-time vs. inference-time positional optimization (454M, L_{train}=2048, 10% passkey mix; 4×/6×/8× extrapolation). YaRN uses the *same fixed* scale s=8 on Geo and EVQ—a matched-scale control, not a YaRN-tuning assessment. PK@8K is 3-seed mean±std; other PK columns and PPL are 3-seed means (per-seed PPL not preserved locally for YaRN rows). PPL uses full-sequence scoring; per-document scoring in Table 25.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	$41_{\pm 5}$%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	$61_{\pm 3}$%	59%	51%	82.9 / 157.7
EVQ	train	3	$53_{\pm 8}$%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**$100_{\pm 0}$%**	**79%**	**68%**	**70.9 / 107.5**

Table 5: PE-dominant summary (125M, L_{train}=128, FineWeb-Edu): in the tested DAPE-style 128→8K protocol, EVQ's closed-form allocation (zero learned parameters) closes more of the extrapolation gap than the learned positional-operator baseline. Geo, DAPE, and EVQ rows are single-seed (seed 42); the Learnable τ row is 3-seed (42/137/256), with mean ± std reported. An L=256 EVQ4+YaRN comparison is in Appendix D.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**$181.2_{\pm 1.3}$**	$437.9_{\pm 12.2}$	−14.8%
DAPE	32	183.6	455.3	−11.4%
EVQ	0	182.0	**333.7**	**−35.0%**

the gap than a 32-parameter learned positional operator on the tested DAPE-style protocol. A complementary τ-sweep at L_{train}=256 (Appendix Fig. 10, numbers in Appendix Table 23) supports that the formula's prediction $\tau^*=d_{\text{head}}/\sqrt{L}$=4.0 is favored among tested settings at both 125M and 454M.

4.4 Primary III: scarce-channel stress test (MLA)

Scarce-channel stress test. MLA [DeepSeek-AI, 2024a,b] compresses RoPE to d_{rope}=32 (16 channels after pairing) vs. 64 in MHA at d_{head}=128; the finite-budget view predicts EVQ's relative effect should *grow* here. On a 432M MLA transformer trained for 500M tokens at L_{train}=8192 (3 seeds), EVQ reduces 2× extrapolation PPL from 138.8 to 95.6 (−31.1%) with +1.1% in-distribution cost, exceeding Geo+YaRN at matched YaRN scale s=4 (117.9; Appendix C.1). The 31.1% MLA effect is larger than the −13.3% MHA improvement at the matched 3-seed FineWeb-Edu setting (Table 21, 16K column), consistent with the hypothesis that scarce channels amplify the allocation correction.

Training-saturation test (preliminary). A standard objection is that standalone wins could be undertraining artifacts. Extending to 1B tokens at L_{train}=4K (single seed): EVQ's raw 2× advantage reverses (−31.1%→+11.1%), but the YaRN composition boost *grows* from +8.6 pp at 500M (L_{train}=8K, EVQ+YaRN inf.) to +13.6 pp at 1B (EVQ+YaRN+FT, −2.5% vs. Geo+YaRN+FT)—the opposite of the undertraining-artifact signature. Preliminary single-seed evidence that the training-time substrate remains the binding constraint on inference-time scaling.

4.5 Robustness and supporting evidence

Multi-scale raw extrapolation (Table 21): in-range PPL stays within ±1.7% while long-range improves 10–46%; capability checks (Table 25) show EVQ is non-destructive. Supporting (1–2 seed): video DiT (0.53× post-hoc), LoRA into LLaMA-3-8B at $r=d_{\text{head}}/2$ (8–19× extrapolation-PPL), progressive training. Aggregated, EVQ's effect attenuates through abstraction layers (raw PPL −46%/−13% 1-/3-seed; gold-NLL −30%; passkey +59 pp; QA +2.2 pp), concentrated at the PE layer. *Takeaway:* RoPE allocation *shape* is a third design axis; EVQ-Cosh ($\tau^*=d_{\text{head}}/\sqrt{L}$) is a closed-form zero-learned-parameter substrate. Production MLA, ≥1B training, and per-channel head-to-heads remain follow-ups.

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A Proof Details

A.1 Collision-only surrogate stationary density

This subsection proves the main-text Theorem 1 (Surrogate stationary allocation) directly from the collision-only surrogate functional; the forced Fisher extension is treated separately in §A.2. Starting from the formal surrogate kernel

$$K_{\text{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{5}$$

the main-text collision functional is (1), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha\rho(\phi) + \beta\, g(\phi) + \nu = 0, \qquad g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi. \tag{6}$$

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\, d\psi, \qquad\qquad g''(\phi) = -\rho(\phi). \tag{7}$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2\rho(\phi) = 0, \qquad \tau = \sqrt{\beta/\alpha}. \tag{8}$$

Evaluating the pre-differentiated stationarity together with $g(0)$=0, $g'(1)$=0, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \qquad \rho'(1) = 0, \tag{9}$$

since differentiating $\alpha\rho(\phi)+\beta g(\phi)+\nu = 0$ once and evaluating at ϕ=1 gives $\alpha\rho'(1)+\beta g'(1) = 0$ hence $\rho'(1)$=0, and at ϕ=0 gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi)$ into (8) and (9) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) \;=\; \frac{\tau\,\cosh(\tau(1-\phi))}{\sinh\tau}, \tag{10}$$

which is main-text Theorem 1. The mass constraint $\int\rho$=1 is not lost: it enters the boundary condition at ϕ=0 via $g'(0)$=$\int_0^1 \rho$=1. Positivity holds on the closed interval since $\rho_\tau(0)$=$\tau\coth\tau$>0 and $\rho_\tau(1)$=$\tau/\sinh\tau$>0 for τ>0. For the degenerate case β=0 the ODE reduces to ρ''=0 with $\rho'(0)$=$\rho'(1)$=0 and $\int\rho$=1, forcing $\rho\equiv 1$, which is also the $\tau\to 0$ limit of (10).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For α>0 and $\beta\geq$0, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha\int\rho^2$ is manifestly PSD on L^2, and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi)\, f(\psi)\, \min(\phi,\psi)\, d\phi\, d\psi \;=\; \int_0^1 \left(\int_s^1 f(u)\, du\right)^2 ds \;\geq\; 0. \tag{11}$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0,1]) \supset C^2([0,1])$ with no further regularity assumption needed for the δ-component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

A.2 Forced Fisher extension

Adding a Fisher utility $-\mu\int\rho(\phi)\, b^{-2\phi}\, d\phi$ to $\mathcal{C}_{\text{app}}$ modifies the constrained stationarity to $\alpha\rho + \beta g - \mu b^{-2\phi} + \nu = 0$, whose twice-differentiated form is

$$\rho''(\phi) - \tau^2\rho(\phi) = \gamma\, b^{-2\phi}, \tag{12}$$

with γ set by the Fisher multiplier's sign convention. The non-resonant general solution ($\tau \neq 2\log b$) is

$$\rho_{\text{full}}(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + \frac{\gamma}{(2\log b)^2 - \tau^2}\, b^{-2\phi}. \tag{13}$$

EVQ-Cosh is the γ=0 homogeneous branch. The forced term is a controlled but nonzero residual; its magnitude and the corresponding warp-amplification caveat are bounded in §A.15.

A.3 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + P\, b^{-2\phi}. \tag{14}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^\phi \rho^*(s)\, ds, \tag{15}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\sinh\tau)\,. \tag{16}$$

To recover geometric RoPE, expand $\sinh\tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{17}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.4 Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether $(\cosh, \sinh)$ and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) The forcing branch is globally small in L^1/CDF transport, not pointwise. Once $\gamma(b) = (\lambda_F/\alpha)\,\bar{\eta}_F\,(2\log b)^2$ is derived (Appendix A.15, Eq. (43)), the $(s^2{-}\tau^2)^{-1}$ denominator in the particular solution $\delta\rho_F^{\text{part}} = \gamma(b)\, q_s/(s^2 - \tau^2)$ is cancelled by the s^2 in $\gamma(b)$, so the forcing amplitude is *not* pointwise $O(1/\log^2 b)$. The true suppression is boundary-layer / L^1-level: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$, giving $\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 = O(1/\log b)$ and CDF-level error of the same order (Eq. (45)). The full solution improves over pure tether by 2–6% *on the offline collision/utility surrogate objective at unit-convention* λ=1 (vs. 24–92% improvement over geometric on the same objective)—this is a surrogate gap, not a trained-PPL claim, and no main empirical claim relies on the forced solution. Furthermore, since inverse-CDF inversion can amplify CDF perturbations by $1/\inf \rho_\tau \sim \sinh\tau/\tau$, at large τ (e.g. τ=4) the forced branch is best treated as a numerical extension rather than dismissed by the L^1 bound; the deployed pure-tether warp captures the dominant Geo→EVQ direction (24–92%) while remaining closed-form. The 99-run Phase 16 sweep (Figure 4) is consistent with the pure-tether family matching the scaling-law prediction, subsuming geometric RoPE as its τ→0 limit.

A.5 Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory's sole approximation. We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) = \int D(\Delta)\cos(\omega(\phi_1)\Delta)\cos(\omega(\phi_2)\Delta)\, d\Delta$ — which is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K? The 24–92% reduction reported below also serves as the operational forced-branch residual diagnostic: if the homogeneous (pure-tether) branch missed the dominant collision-reduction direction, the exact-kernel test would show no consistent improvement at the deployed τ.

Table 6 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in \{128, \ldots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in \{16, 32\}$), covering both text and video regimes.

Table 6: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L, larger b).

	Collision C		Eff. rank	
Configuration	GEO	EVQ (Δ_C)	GEO	EVQ
Text: K=32, b=500K, *sweep* L				
L=128	226.3	17.7 (−92%)	2.5	17.0
L=256	192.6	27.6 (−86%)	3.0	13.9
L=512	163.2	43.3 (−73%)	3.6	10.8
L=1024	133.9	59.0 (−56%)	4.4	8.8
L=2048	109.0	67.5 (−38%)	5.4	8.0
L=4096	87.0	66.3 (−24%)	6.6	8.1
Text: K=32, L=2048, *sweep* b				
b=10K	34.8	19.2 (−45%)	13.0	17.3
b=100K	79.4	47.0 (−41%)	7.1	10.4
Video: K=16, L=32, *sweep* b				
b=100	22.6	14.2 (−37%)	4.8	6.4
b=1K	42.3	28.0 (−34%)	2.8	3.9
b=10K	55.4	39.8 (−28%)	2.2	2.9
b=50K	63.3	47.2 (−26%)	2.0	2.5

EVQ reduces collision under the exact kernel in all 12 configurations (−24% to −92%), with effective rank improvements of +24% to +570%. Two patterns emerge. First, the collision reduction monotonically increases as the geometric allocation becomes more degenerate: shorter L (more extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE (L=4096, where baseline effective rank is already 6.6), EVQ still improves collision by −24%, supporting the interpretation that the surrogate identifies a genuine structural improvement within the tested parameter range, not only in extreme regimes.

A.6 Exact higher-order displacement floor

The one-channel mid-displacement threshold admits an exact Taylor series in $1/K$ that is fully closed-form (no empirical calibration).

Proposition 2 (Higher-order floor)**.** *For the EVQ-Cosh mapping with $K = d_{\mathrm{rot}}/2$ channels, the N-channel mid-displacement threshold satisfies*

$$\tau_{\mathrm{floor}}(N,K) \;=\; 4\sqrt{\frac{N}{K}}\left[1+\frac{N}{2K}+\frac{241}{120}\left(\frac{N}{K}\right)^2+O(K^{-3})\right]. \tag{18}$$

All coefficients are derived from the Taylor series $\Delta\varphi(1/2,\tau) = -\tau^2/16 + \tau^4/256 + 17\tau^6/30720 + O(\tau^8)$, whose coefficients $(-1/16, +1/256, +17/30720)$ are exact rationals from the expansion of $\operatorname{arcsinh}(\sinh\tau/2)/\tau$ *(Appendix script* `theory_B_floor_higher_order.py`*; verified symbolically).*

Accuracy. Against the exact floor (obtained by numerically solving $|\Delta\varphi(1/2,\tau)| = N/K$): the leading term $4\sqrt{N/K}$ has relative error 3.85% at K=16 and 0.20% at K=512; the two-term expansion reduces this to 0.088% at K=16 and below 10^{-5} for $K\geq128$. The formula is thus a closed-form refinement of the Habitable Zone lower bound, with all constants derived from the cosh geometry alone.

A.7 Waterbed envelope and corollary

Setup. With $\rho>0$, $\int_0^1 \rho=1$, $U(d\phi)=d\phi$, $P_\rho(d\phi)=\rho\,d\phi$, $\delta\rho=\rho-1$, and $M_\rho(s)=\int_s^1 \delta\rho\,du$, define the waterbed functional

$$\mathcal{W}(\rho) \;\coloneqq\; \alpha\,\chi^2(U\|P_\rho) + \beta\int_0^1 M_\rho(s)^2\,ds, \qquad \chi^2(U\|P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho}\,d\phi. \tag{19}$$

The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the integrated squared cumulative error, which arises from the min-kernel off-diagonal covariance in K_{app}.

Proposition 3 (Waterbed inequality)**.** *Let $D_B(\rho) = -\int_0^1 \log\rho(\phi)\,d\phi$ be the Burg entropy deficit. Then*

$$\mathcal{W}(\rho) \;\geq\; \alpha\,F_*(D_B(\rho)), \qquad F_*(d) \;=\; \inf_{\rho>0,\,\int\rho=1,\,-\int\log\rho=d}\left[\int_0^1 \frac{1}{\rho}-1\right], \tag{20}$$

with equality $\mathcal{W}(\rho)=0$ if and only if $\rho\equiv1$ (geometric RoPE). A readable corollary via Rényi monotonicity is

$$\mathcal{W}(\rho) \;\geq\; \alpha\big(e^{D_B(\rho)}-1\big). \tag{21}$$

Proof sketch. Since $\beta\int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha\chi^2(U\|P_\rho)$. For fixed $D_B(\rho)$=d, the sharp lower envelope of χ^2 over all densities is the Csiszár–Vajda f-divergence comparison F_*, whose extremizer is a two-level density $X = 1/\rho \in \{a,c\}$ with mass-normalization and entropy-deficit constraints. For the corollary, $1+\chi^2(U\|P_\rho) = \int 1/\rho\,\mathbb{E}_U[\cdot] \geq e^{D_1(U\|P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality at $\rho\equiv1$ is immediate from both sides vanishing. □

EVQ evaluation. For $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$ (Eq. (23) below uses the unnormalized integral; cf. §A.16 "Normalization convention"),

$$D_B(\rho_\tau) \;=\; \log\tfrac{\sinh\tau}{\tau} - \tfrac{1}{\tau}\int_0^\tau \log\cosh y\,dy \;=\; \tfrac{\tau^4}{90} - \tfrac{8\tau^6}{2835} + O(\tau^8), \tag{22}$$

$$S_{\chi^2}(\rho_\tau) \;=\; \tfrac{\sinh\tau}{\tau^2}\arctan(\sinh\tau) - 1 \;=\; \tfrac{\tau^4}{45} - \tfrac{2\tau^6}{315} + O(\tau^8), \tag{23}$$

$$\int_0^1 M_{\rho_\tau}(s)^2\,ds \;=\; \tfrac{2\tau^4}{945} + O(\tau^6). \tag{24}$$

Combining,

$$\mathcal{W}(\rho_\tau) \;=\; \left(\frac{\alpha}{45}+\frac{2\beta}{945}\right)\tau^4 + O(\tau^6), \qquad \mathcal{W}(\rho_{\tau_*}) \;=\; O\!\left(\frac{d_{\mathrm{head}}^4}{(L_{\mathrm{eff}}^J)^2}\right). \tag{25}$$

The in-range surrogate cost for EVQ is $O(\tau^4)$; since $\tau_*^2 \asymp d_{\mathrm{head}}^2/L_{\mathrm{eff}}^J$, the cost scales as $d_{\mathrm{head}}^4/(L_{\mathrm{eff}}^J)^2$. This is the explicit form of "bounded in-range cost" cited in the main text.

A.8 Surrogate self-consistency theorem

The surrogate functional $\mathcal{J}[\rho]$ admits three exact identities that connect its two integral terms $T_1(\tau) = \int_0^1 \rho^2\, d\phi$ and $T_2(\tau) = \int_0^1\!\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\, d\phi\, d\psi$, where $\rho(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$ is the normalized cosh density.

Theorem 3 (Surrogate self-consistency)**.** *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau\coth\tau. \tag{26}$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2\rho$ this gives $h'' = -\tau^2\rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2\rho - (-\tau^2\rho) = 0$, so f is linear in ϕ. The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \tag{27}$$

Integrating against ρ:

$$T_2 = \int_0^1 \rho(\phi)\, g(\phi)\, d\phi = \frac{1}{\tau^2}\big[\rho(0)\underbrace{\int_0^1 \rho}_{=1} - T_1\big] = \frac{\rho(0) - T_1}{\tau^2}. \tag{28}$$

Since $\rho(0) = \tau\cosh\tau/\sinh\tau = \tau\coth\tau$, rearranging gives (26). □

An immediate corollary follows.

Corollary 1 (Closed-form T_2)**.**

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau\sinh^2\tau}. \tag{29}$$

This follows from substituting the known closed form $T_1(\tau) = \tau(1 + \sinh 2\tau/2\tau)/(2\sinh^2\tau)$ and $\rho(0) = \tau\coth\tau$ into the theorem. Both identities are verified numerically to machine precision ($< 10^{-15}$ relative error across $\tau \in [0.01, 20]$; code: `scripts/lambda_identity.py`).

The self-consistency theorem reveals that the surrogate's two energy terms are not independent: they are locked by the cosh density's self-reproducing property under the min-kernel Green's function. This structural rigidity is why the pure-tether solution (§A) performs so well despite being a single-parameter family.

A.9 Cross-validation of λ

The scaling law $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$ is analyzed at two levels.

Collision-score level. We minimize the exact-kernel collision score $C(\tau)$ for each of 9 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, 512, 1024\}$) and compute $c_{\text{coll}} = \tau^*_{\text{coll}}/(d_{\text{head}}/\sqrt{L})$. The resulting $c_{\text{coll}} \approx 1.17$ is nearly constant (CV: 0.11%); leave-one-out cross-validation gives max prediction error $< 0.4\%$ (Table 7). This near-zero spread reflects the smoothness of the collision score as a function of τ, not PPL variability.

PPL-sweep level. Phase 16 training sweeps (9 configurations $\times$ 3+ seeds) identify the best observed PPL sweep point for each configuration. The resulting $\lambda_{\text{PPL}} = \tau^*_{\text{PPL}}/(d_{\text{head}}/\sqrt{L})$ has mean $\approx$1.19 with broader scatter (CV $\approx$14%), because PPL depends on training noise, finite data, and the discrete sweep grid. The deployed formula uses λ=1 (i.e. $\tau = d_{\text{head}}/\sqrt{L}$ without a multiplicative constant), which consistently falls inside the flat PPL basin: the PPL gap between the formula and the best observed grid point is <1% across all 27 tested configurations.

Table 7: Collision-score validation of the operating-point prefactor c in $\tau^* = c \cdot d_{\text{head}}/\sqrt{L}$. Column $c_{\text{pred}}(L,b) = \sqrt{45\,Q_1(L,b)}$ is the theory prediction under the λ=1 unit convention (Appendix A.17); column $c_{\text{coll}} = \tau^*_{\text{coll}}/(d_{\text{head}}/\sqrt{L})$ is the exact-kernel collision-score minimizer. The ratio $c_{\text{coll}}/c_{\text{pred}} = 0.981 \pm 0.003$ (CV 0.28%, 9 configurations) is near-unity, supporting surrogate–exact curvature agreement at the 2% level; implied curvature ratio $\lambda_\infty = 0.96 \pm 0.005$. Leave-one-out cross-validation on c_{coll} gives max prediction error 0.008 ($<$ 0.4% relative, RMSE 0.004), so the near-zero CV reflects the smoothness of the collision score, not PPL variability (PPL sweep best points exhibit CV ≈14%). The deployed formula uses λ=1, which falls inside the $<$ 1% PPL basin across all 27 configurations.

d_{head}	L	Q_1	c_{pred}	c_{coll}	Ratio	Error
32	256	0.03145	1.190	1.170	0.984	+0.002
32	512	0.03192	1.199	1.174	0.979	−0.005
32	1024	0.03159	1.192	1.170	0.981	+0.001
64	256	0.03145	1.190	1.170	0.984	+0.004
64	512	0.03192	1.199	1.170	0.976	+0.002
64	1024	0.03159	1.192	1.170	0.981	+0.002
128	256	0.03145	1.190	1.171	0.985	−0.004
128	512	0.03192	1.199	1.172	0.978	−0.008
128	1024	0.03159	1.192	1.170	0.981	+0.004
Mean $c_{\text{coll}}/c_{\text{pred}} = 0.981 \pm 0.003$ (CV 0.28%); implied $\lambda_\infty = 0.96 \pm 0.005$						

Basin flatness. The collision-score landscape $C(\tau)$ for a representative configuration (d_{head}=64, L=512, b=500K) is flat near the optimum: ±20% deviation from τ^*_{coll} incurs <1% collision increase. This flatness is why the deployed bare formula $\tau = d_{\text{head}}/\sqrt{L}$ (i.e. omitting the $\sqrt{45\,Q_1(L,b)}$ prefactor for practical simplicity, so c_{deployed}=1) sits ≈15% below c_{coll}=1.171—inside the ±20% collision basin—and consequently inside the <1% PPL basin observed in trained sweeps. The full theoretical prediction $c_{\text{pred}}=\sqrt{45\,Q_1}\approx 1.19$ matches c_{coll} to 2% (see "Numerical validation" below); the practical and theoretical prefactors differ, but both land inside the same flat basin.

b-dependence of c_{pred}. The basin width above is verified at the representative (d_{head}, L, b)=(64, 512, 500K). The theoretical prefactor $c_{\text{pred}}(L,b)=\sqrt{45\,Q_1(L,b)}$ varies modestly with b: at b=500K across $L\in[128, 4096]$, $c_{\text{pred}}\in[1.14, 1.20]$ and the deployed c=1 deviates 12–17% (inside the ±20% collision basin); at b=100K the range is [1.06, 1.20] and deployed c=1 stays inside the basin throughout; at b=10K, c_{pred} drops to 0.94 at L=2048 (+7% deviation, still inside the basin) and 0.80 at L=4096 (+25% deviation, marginally *outside* the ±20% basin). All primary experiments use $b\geq$500K where the bare formula stays inside the basin; at small b and large L, reverting to the explicit $c_{\text{pred}}(L,b)=\sqrt{45\,Q_1(L,b)}$ prefactor is appropriate.

Why λ is a normalization convention, not a free parameter. The Lagrange multiplier λ in the softmax-transport functional $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda\,U(\tau, L)$ has a closed-form interpretation. We use d_{head} (not the number of RoPE frequency pairs $K = d_{\text{head}}/2$) as the stiffness normalization throughout both main text and appendix; all factors of two arising from the RoPE pair structure are absorbed into the $\mathcal{O}(1)$ convention-dependent prefactor. We adopt the Dirichlet-energy convention in which S_{χ^2} enters $\mathcal{F}$ with a $\frac{1}{2}$ prefactor, so that under the small-τ expansion $\frac{1}{2}S_{\chi^2}(\tau) = \tau^4/(90\,d_{\text{head}}) + O(\tau^6)$ (using $S_{\chi^2}(\tau) = \tau^4/(45\,d_{\text{head}})$ from §A.11); this matches the numeric constant 45 appearing in Proposition 1. Similarly, $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L,b)] + O(\tau^4)$, where $Q_1(L,b) = \int_0^1 \eta(\phi)\,q(Lb^{-\phi})\,d\phi$ is an explicit integral of known functions. All numerical factors below are reported under this $\frac{1}{2}S$ convention; a reader using an un-normalized S would absorb a factor of 2 into λ. The balance equation $\partial\mathcal{F}/\partial\tau = 0$ yields

$$\tau_*^2 \;=\; 45\,\lambda\,Q_1(L,b)\cdot\frac{d_{\text{head}}^2}{L}.$$

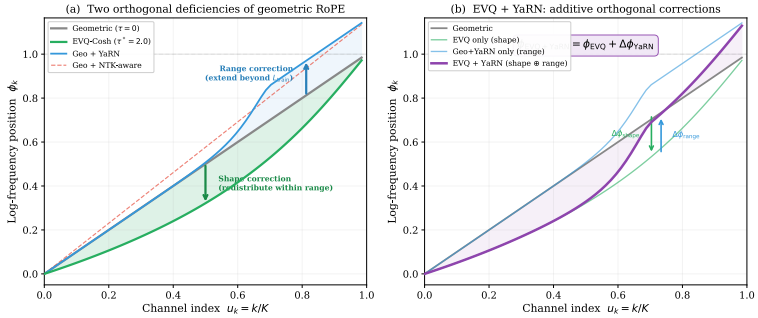

Figure 3: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** EVQ bends the allocation *below* the diagonal (shape); YaRN shifts it *above* (range). Their combination is additive. Setup: base = 10,000, d_{head}=128, L_{train}=4096, 4× extrapolation.

Splitting Q_1 into an active-channel contribution ($q \approx 1/2$ for $Lb^{-\phi} \gg 1$, i.e. $\phi \in [0, \phi^*]$ with $\phi^* = \ln L / \ln b$) and a dead-channel contribution ($q(x) = x^4/45 + O(x^6)$ for $x = Lb^{-\phi} \ll 1$ on $\phi \in [\phi^*, 1]$; the leading Taylor coefficient is $x^4/45$, not $x^2/45$) gives the closed form $Q_1(L, b) \approx \frac{1}{12}\big[1 - (1 - \phi^*)^3 - \phi^*\big]$ to leading order, with the dead-channel correction exponentially small. To avoid symbol reuse, we reserve λ for the utility-weight normalization in $\mathcal{F}(\tau)=\frac{1}{2}S_{\chi^2}(\tau)-\lambda U(\tau, L)$ and write the multiplicative operating-point prefactor as c, so $\tau^* = c \cdot d_{\text{head}}/\sqrt{L}$. Combining the expansions above gives the theoretical prefactor $c_{\text{pred}}(L, b; \lambda) = \sqrt{45\,\lambda\, Q_1(L, b)}$. Under the deployed convention λ=1, this is a posterior-consistency prediction (computable from (L, b)), not a per-experiment fitted hyperparameter; the residual $\mathcal{O}(1)$ offset is absorbed into the flat PPL basin rather than treated as a tunable parameter.

Probabilistic interpretation of $q(x)$. The function $q(x)=1/2+\sin(2x)/(4x)-(\sin x/x)^2$ is not a postulated form; it is the cosine phase-variance under a uniform separation prior $t \sim \text{Unif}[0, 1]$:

$$q(x) \;=\; \text{Var}_{t\sim U[0,1]}\big[\cos(xt)\big] \;=\; \mathbb{E}_t\big[\cos^2(xt)\big] - \big(\mathbb{E}_t\big[\cos(xt)\big]\big)^2,$$

since $\mathbb{E}_t[\cos^2(xt)] = 1/2 + \sin(2x)/(4x)$ and $\mathbb{E}_t[\cos(xt)] = \sin(x)/x$. Operationally, $q(Lb^{-\phi})$ measures how much information channel ϕ retains about a uniformly distributed relative position over $[0, L]$: in the dead-channel tail ($Lb^{-\phi} \ll 1$) the cosine is nearly constant and $q \to 0$; in the active region ($Lb^{-\phi} \gg 1$) the phase is well-sampled and $q \to 1/2$. Thus $Q_1(L, b)$ weights the cosh-driven density perturbation $\eta(\phi)$ by the per-channel phase-variance budget, and $q(x)$ is a self-contained signal-processing quantity rather than an unmotivated kernel.

A.10 Scaling analysis of τ^*: from surrogate to operating point

The scaling law

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \tag{30}$$

is supported across 99 trained models spanning 27 validation settings (Figure 4); primary claims rely on the 3-seed anchors within this grid, with 1–2-seed rows extending scope. Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha\delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L}\big[\frac{\sin((\omega_1-\omega_2)L)}{\omega_1-\omega_2} + \frac{\sin((\omega_1+\omega_2)L)}{\omega_1+\omega_2}\big]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \ldots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \;\approx\; \frac{1}{2K} \;=\; \frac{1}{d_{\text{head}}}, \qquad \beta \;\sim\; L^{-0.22},\; O(1). \tag{31}$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L. The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \tag{32}$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the observed scaling. However, the L-scaling is too weak: $L^{-0.11}$ vs. the observed $L^{-0.5}$.

The discrete–continuous gap. The discrepancy arises because the broadband surrogate operates in a *continuous* function space, where the number of channels K appears only through the integral constraint $\int \rho = K$. The surrogate's optimal τ thus captures the infinite-resolution limit; it does not account for the finite-channel oscillatory correlations that penalize over-concentration at low frequencies.

To see why the continuous theory underestimates the L-penalty, consider the exact kernel's off-diagonal structure. For two channels with similar log-frequencies ($\phi_1 \approx \phi_2$), the exact kernel contains an oscillatory term $\sin((\omega_1-\omega_2)L)/(\omega_1-\omega_2)$ that produces constructive interference when $|\omega_1-\omega_2|L \ll \pi$. Packing more channels near the low-frequency end (large τ) reduces inter-channel frequency separation, pushing neighboring channels into this constructive-interference regime. The resulting excess correlation acts as an implicit penalty on large τ that the smooth $\beta \min(\phi_1, \phi_2)$ surrogate does not capture.

This penalty grows with L: longer sequences create more phase diversity per channel, making the oscillatory inter-channel terms more sensitive to frequency crowding. This discrete-correction picture is consistent with the observed $\tau^* = d_{\text{head}}/\sqrt{L}$ scaling: a stronger L-penalty would steepen the surrogate's $L^{-0.11}$ into something closer to $L^{-1/2}$, and a finite-channel correction scaling with $K = d_{\text{head}}/2$ would plausibly strengthen the $\sqrt{d_{\text{head}}}$ factor, but we do not treat this heuristic as a derivation. The rigorous $L^{-1/2}$ recovery is the softmax-transport argument in Proposition 1.

In support of this picture, at the special point $d_{\text{head}} = L$ (where the number of encoding dimensions equals the number of positions and the continuous and discrete problems roughly coincide), the surrogate formula $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$ matches $d_{\text{head}}/\sqrt{L}$ to within 3%; as L/d_{head} grows, the surrogate progressively overestimates τ^*, consistent with a non-negligible discrete-correction term outside its regime of validity.

Summary. The broadband surrogate supplies two of the three ingredients for τ^*: the *functional form* (cosh density, via the ODE) and the d_{head} *dependence* (via $\alpha = 1/d_{\text{head}}$). The $L^{-1/2}$ exponent is obtained from the softmax transport analysis (Proposition 1 and §A.11): the post-softmax Jacobian introduces a $1/L$ factor per channel, and combining this with Pearson χ^2 stiffness yields $\gamma = 0.465$ (Table 9), which we round to $\gamma \approx 0.5$ for the practical rule. Together, these yield

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}}, \tag{33}$$

whose observed best-τ points are highly correlated with the deployed rule within the tested grid (Figure 4).

Configuration breakdown. Table 8 decomposes the 27 tested configurations by architecture, seed count, and evidence tier.

A.11 Stiffness functional derivation

The softmax transport objective in Proposition 1 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family

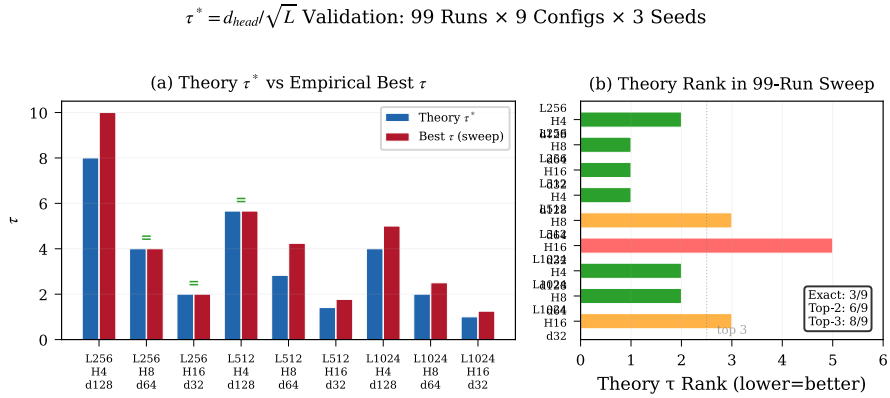

Figure 4: **Scaling-law sweep over 99 trained models across 27 validation settings.** Each point is a trained model from the Phase-16 3-seed grid or a scoped supporting validation row (detailed in Table 7 and Appendix D); the x-axis is the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ and the y-axis is the best observed sweep point for τ under validation perplexity. Primary claims rely only on the 3-seed primary settings; 1–2-seed rows extend the tested grid but do not anchor the main empirical claims.

Table 8: Breakdown of the 27 configurations in the tested grid for $\tau^* = d_{\text{head}}/\sqrt{L}$. Primary claims rely on the 3-seed primary rows; 1–2-seed rows extend scope. The 0.53× modality correction for video DiT is empirical (§B.5).

Architecture	Scale	d_{head}	L_{train}	Seeds	τ^* variant	Tier
MHA (text)	50M	64	2048	1	base	Robustness
MHA (text)	125M	64	128	1–3	base	Primary
MHA (text)	125M	64	2048	1	base	Robustness
MHA (text)	454M	64	128	1–3	base	Primary
MHA (text)	454M	64	256	3	base	Primary
MHA (text)	454M	64	512–2048	3	base	Primary
MHA (text)	454M	64	2048	3	base	Primary
MHA (text)	750M	64	4096	1	base	Supporting
MLA	432M	128	8192	3	base	Primary
DiT (video)	129M	16	32	2	0.53×	Supporting
DiT (video)	382M	16	32	1	0.53×	Supporting
LoRA	8B	128	8192	1	base	Supporting

$S_p(\tau) = \frac{1}{d_{\text{head}}} \int_0^1 (\rho_\tau - 1)^2/\rho_\tau^p \, d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$.

Why χ^2. Two independent arguments favor p=1. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int (\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2.

Closed form. For p=1, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \cdot \arctan(\sinh \tau)}{\tau^2} - 1 \right], \tag{34}$$

verified numerically to relative error $<7\times10^{-6}$.

Exponent-matched diagnostic. As a sensitivity analysis, one can ask which member of the S_p family would match the target exponent $\tau_* \propto 1/\sqrt{L}$ exactly over this grid. The

Table 9: L-exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Exponent-matched diagnostic	0.80	0.498	Sensitivity analysis, not used in deployment
Pearson χ^2	**1.0**	**0.465**	Channel-load variance penalty (axioms)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi (α=2)	—	0.807	

resulting $S_{\text{derived}}(\tau)$ lies near $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), close to but distinct from the Pearson χ^2 choice p=1. We do not use this as a derivation of the stiffness; it is reported as sensitivity analysis. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ∼60s).

A.12 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. This subsection gives a phenomenological model calibrated by the rank sweep, not a first-principles derivation: the EVQ warp actually perturbs frequencies at order τ^2 (Eq. (36)), so the τ^2 coupling below is an *effective* single-scale surrogate for the leading non-perturbative interaction with frozen pretrained Q/K blocks, not the literal leading term of the small-τ expansion. Treating the frozen-weight mismatch as producing an aggregate logit variance $\mathrm{Var}(\delta\ell) \propto \tau^2$, the total stiffness becomes

$$S_{\text{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0\Big(1 - \frac{r}{K}\Big)\,\frac{\tau^2}{d_{\text{head}}}, \tag{35}$$

where r is the LoRA rank, $K = d_{\text{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. In this phenomenological model the coupling term dominates at small τ, and the system exhibits a *near-phase-transition* in effective viability: $\tau_* \to 0$ for small r/K and τ_* jumps to an $\mathcal{O}(1)$ value once r is large enough.

Numerical verification under this phenomenological model shows the transition is *apparent-sharp*: for LLaMA-8B parameters (d_{head}=128, K=64), sweeping LoRA rank over the variational landscape shows τ_*=0 for all $r \leq 48$ ($r/K \leq 0.75$), then τ_* jumps to 1.41 at r=64 (r/K=1.0). The rank at which this occurs, $r \approx K$, is the value observed in this sweep; we do not claim an exact first-principles derivation of $r_c = K$, and the specific threshold depends on Λ_0.

For LLaMA-8B with LoRA r=16: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At τ=1.414, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (this PPL value is reproduced after calibrating Λ_0 on the same observation, so it is a calibration check of the phenomenological model rather than an out-of-sample prediction). The qualitative implication—LoRA with $r \gtrsim K$ should restore EVQ viability—is independent of this calibration and is consistent with the rank sweep in Table 24.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau^*_{\text{SFT}}(T) \approx \tau^*_{\text{pt}}/\sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ^*_{pt} requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

A.13 Mechanism isolation: collision-only ablation

To verify that the collision-reduction mechanism identified by the surrogate theory is the primary driver of EVQ's improvement, we perform a diagnostic experiment that isolates the collision objective from the full training loss.

We define the off-diagonal energy $E_{\text{off}} = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ under the exact RoPE kernel (not the surrogate), and optimize the EVQ allocation to minimize E_{off} directly ("collision-only"), without any language modeling. Table 10 compares geometric, EVQ at theory-predicted τ^*, and collision-optimized allocations across 36 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, \ldots, 8192\}$, $b \in \{10\text{K}, 500\text{K}\}$).

Table 10: Mechanism isolation: off-diagonal energy E_{off} and mutual coherence μ across representative configurations. Column "Oracle" reports a numerical search (collision-only objective, not a global minimizer; see caveat below). Geo and EVQ columns are evaluated at fixed allocations; −65% annotations are computed against the Geo baseline.

	E_{off} (↓)			Mutual coherence μ	
Configuration	GEO	EVQ(τ^*)	Oracle	GEO	Oracle
b=10K					
d=32, L=512	653	516	220 (−66%)	.999	.708
d=64, L=512	2887	1533	1000 (−65%)	.999	.814
d=128, L=512	12120	3958	4283 (−65%)	.999	.973
b=500K					
d=32, L=512	850	690	286 (−66%)	1.00	.862
d=64, L=512	3715	2069	1305 (−65%)	1.00	.995
d=128, L=2048	16473	8899	5574 (−66%)	1.00	.998

Two consistent findings emerge across configurations, plus one caveat. First, both EVQ and the oracle search reduce E_{off} substantially relative to geometric in all tested configurations, and mutual coherence μ drops from ≥0.99 under geometric to 0.55–0.99 under the oracle—EVQ breaks the near-degeneracy of the geometric kernel in the same direction the oracle does. Second, the oracle capture fraction (defined as $(E_{\text{Geo}} - E_{\text{EVQ}})/(E_{\text{Geo}} - E_{\text{Oracle}})$) varies across configurations, so we report the individual EVQ and oracle columns rather than a single capture statistic. *Caveat on the oracle column.* The oracle is a numerical collision-only search and is not guaranteed to find the global minimum; in the d=128, L=512, b=10K row, EVQ at the theory-predicted τ^* achieves a lower E_{off} than the oracle search, indicating the search missed the true minimum at that configuration. We read the oracle numbers as a directional lower bound rather than as a tight global optimum.

A.14 Effective window L_{eff}^J via softmax-Jacobian Rayleigh quotient

From density perturbation to RoPE frequency shift. The EVQ pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh\tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1}\operatorname{arsinh}((1-u)\sinh\tau)$. Setting $\theta=\tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta\, a(\phi) + O(\theta^2)$ with $a(\phi) = (1-\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) \;=\; u + \tau^2\, h(u) + O(\tau^4), \qquad h(u) \;=\; -\int_0^u a(v)\, dv \;=\; -\tfrac{1}{6}u(1-u)(2-u), \tag{36}$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)}[1 - \tau^2(\log b)\, h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)}=b^{-u_k}$. Hence $\dot\omega_k \coloneqq \partial_\theta \omega_k|_{\tau=0} = -(\log b)\, \omega_k^{(0)} h(u_k)$.

Explicit RoPE logit transport direction. Fix a layer/head and query i; write each 2D RoPE pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1} + iq_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1} + ik_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}}\overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} + iD_{ij,k}$. With relative position r_{ij} and attention scale s_{att}, the RoPE logit is $z_{ij}^{\text{rope}}(\tau) = s_{\text{att}}^{-1}\operatorname{Re}\sum_k \alpha_{ij,k}\exp(ir_{ij}\omega_k(\tau))$. The leading logit transport direction $g_i(j) \coloneqq \partial_\theta z_{ij}^{\text{rope}}|_{\tau=0}$ has the closed form

$$g_i(j) \;=\; \frac{(\log b)\, r_{ij}}{s_{\text{att}}} \sum_{k=1}^{K} \omega_k^{(0)}\, h(u_k)\Big[\, C_{ij,k}\sin(r_{ij}\omega_k^{(0)}) + D_{ij,k}\cos(r_{ij}\omega_k^{(0)})\,\Big]. \tag{37}$$

This is the Fréchet derivative of the logit with respect to the EVQ allocation perturbation at fixed activations. A finite-τ implementation that avoids first-order phase expansion at large

$r_{ij}\omega$ is

$$g_i^{(\tau)}(j) \;=\; \frac{1}{s_{\mathrm{att}}\tau^2}\,\mathrm{Re}\sum_{k=1}^{K}\alpha_{ij,k}\Big[\exp(ir_{ij}b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij}b^{-u_k})\Big], \tag{38}$$

with $g_i(j) = \lim_{\tau\to 0} g_i^{(\tau)}(j)$.

Primary definition: Rayleigh quotient of the softmax Jacobian. For the trained attention distribution $p_i(j)$ on keys $j\in\mathcal{K}_i$, $J_{\mathrm{sm}}(p_i) = \mathrm{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\mathrm{sm}}(p_i) g_i = \mathrm{Var}_{j\sim p_i}[g_i(j)] = \frac{1}{2}\sum_{j,k} p_i(j)p_i(k)(g_i(j)-g_i(k))^2$. Since $J_{\mathrm{sm}}(p_i)\mathbf{1}{=}0$, we center g_i by the key-wise mean operator P_i. The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\mathrm{eff}}^J} \;=\; \kappa_{\mathrm{att}} \;=\; \frac{\mathbb{E}_{\ell,h,i}\big[\, g_i^\top J_{\mathrm{sm}}(p_i) g_i \,\big]}{\mathbb{E}_{\ell,h,i}\big[\, \|P_i g_i\|_2^2 \,\big]}, \tag{39}$$

which is the directional post-softmax curvature in the RoPE transport direction, averaged with the natural utility weight $\|P_i g_i\|_2^2$. We prefer (39) over an average of per-query ratios because it avoids noise amplification when $\|P_i g_i\|_2$ is small. Substituting $1/L$ by $1/L_{\mathrm{eff}}^J$ in Proposition 1 gives

$$\tau_*^2 \;=\; 45\,\lambda\, Q_1^{\mathrm{eff}}(L,b)\,\frac{d_{\mathrm{head}}^2}{L_{\mathrm{eff}}^J}, \qquad \tau^* \approx \frac{d_{\mathrm{head}}}{\sqrt{L_{\mathrm{eff}}^J}},$$
$$\frac{d\log\tau^*}{d\log L_{\mathrm{train}}} \;=\; -\tfrac{1}{2}\,a(L_{\mathrm{train}}), \qquad a \;\coloneqq\; \frac{d\log L_{\mathrm{eff}}^J}{d\log L_{\mathrm{train}}}. \tag{40}$$

The paper's $-1/2$ exponent is the case $a{=}1$ (diffuse growth); $a{\to}0$ corresponds to saturation $L_{\mathrm{eff}}^J \to L_* < \infty$.

Entropy proxy and its limits. A cheap diagnostic is the harmonic mean of per-query entropy lengths e^{H_i}, where $H_i = -\sum_j p_i(j)\log p_i(j)$:

$$\frac{1}{L_{\mathrm{eff}}^H} \;=\; \mathbb{E}_{\ell,h,i}\big[\, e^{-H_i} \,\big]. \tag{41}$$

If each active window is approximately flat on support W_i of size m_i and g_i has energy concentrated on W_i, then $e^{-H_i}{=}1/m_i$ and $g_i^\top J_{\mathrm{sm}}(p_i) g_i / \|P_{W_i} g_i\|_2^2 = 1/m_i$, giving $L_{\mathrm{eff}}^J = L_{\mathrm{eff}}^H$. The proxy and theoretical object differ in two regimes. First, if attention has nonuniform masses $p_j = (1{+}\varepsilon_j)/m$ and g_i covaries with ε_j, then the Rayleigh quotient picks up the first-order alignment term $\sum_j \varepsilon_j g_j^2 / \sum_j g_j^2$ which e^{-H} lacks. Second, for near-one-hot attention ($H_i{\to}0$), $e^{-H_i}{\to}1$ would suggest "effective length ≈ 1", whereas $J_{\mathrm{sm}}(p_i)\to 0$: the softmax saturates and post-softmax sensitivity vanishes, so L_{eff}^J reads sharp attention as low-transport rather than short-but-sensitive. We therefore treat L_{eff}^J as the theoretical quantity and L_{eff}^H as a cheap flat-window proxy.

Perturbative self-consistency of Proposition 1. The replacement $1/L \mapsto 1/L_{\mathrm{eff}}^J$ in Proposition 1 is not a heuristic substitution but a consequence of a standard softmax Taylor expansion. Writing the log-partition function $A(z) = \log\sum_j e^{z_j}$ and $p_\epsilon = \mathrm{softmax}(z{+}\epsilon g)$, one has $D_{\mathrm{KL}}(p\|p_\epsilon) = A(z+\epsilon g) - A(z) - \epsilon p^\top g = \frac{1}{2}\epsilon^2\, g^\top J_{\mathrm{sm}}(p)\, g + R_3$ with $|R_3| \le C_3|\epsilon|^3\|g\|_\infty^3$. Since the second derivative of A is $J_{\mathrm{sm}}(p) = \mathrm{diag}(p) - pp^\top$, the directional curvature $g^\top J_{\mathrm{sm}}(p) g = \mathrm{Var}_{j\sim p}[g_j]$ depends only on the distribution p. For $p_0 = 1/L$ this reduces to $\|Pg\|^2/L$, the unique origin of the $1/L$ factor in the diffuse-baseline derivation. Substituting $1/L \mapsto \mathbb{E}[g^\top J_{\mathrm{sm}}(p) g]/\mathbb{E}[\|Pg\|^2] = 1/L_{\mathrm{eff}}^J$ and re-running the small-τ variational balance $F(\tau) = S_{\chi^2}(\tau) - \lambda U(\tau,p)$ gives $\tau_*^2 = 45\lambda Q_1^J(p,b)\, d_{\mathrm{head}}^2/L_{\mathrm{eff}}^J + O(d_{\mathrm{head}}^4/(L_{\mathrm{eff}}^J)^2)$, with the residual R_3 absorbed into higher-order terms under the assumption $|\tau^2|\cdot\max_j|g_j|$ small. Proposition 1 thus extends from the diffuse baseline to any trained attention distribution satisfying this smallness condition.

Validation protocol. No retraining is required for the proxy test. On existing 16K or 32K checkpoints, log attention matrices on $\sim 10^3$ sequences and compute (41). For the theoretical object, evaluate (37) or (38) on the same batch and substitute into (39). A falsifying measurement would be either $L^J_{\text{eff}} \approx L_{\text{train}}$ with a saturating empirical optimum, or $L^J_{\text{eff}} \to L_* < \infty$ with $\tau^* \propto L_{\text{train}}^{-1/2}$ continuing to hold at $L_{\text{train}} \geq 16\text{K}$.

A.15 Pure-tether branch: Fisher-forcing coefficient and residual bound

Setup. The full stationary ODE under $\mathcal{C}_{\text{app}}$ plus a Fisher utility is $\rho'' - \tau^2 \rho = \gamma\, b^{-2\phi}$. Writing $s = 2\log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau \cosh(\tau(1-\phi))/\sinh\tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta\rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha\rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi)\, b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) \;=\; \frac{1}{2\, s_{\text{att}}^2}\, \mathbb{E}_{\ell,h,i,j}\big[\, w_{\ell hij}\, |\alpha_{\ell hij,k}|^2 \,\big]\,, \qquad \alpha_{\ell hij,k} \;=\; q^{\mathbb{C}}_{i,k}\, \overline{k^{\mathbb{C}}_{j,k}}. \tag{42}$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2\rho = (\lambda_F/\alpha) V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\boxed{\gamma(b) \;=\; \frac{\lambda_F}{\alpha}\, \bar{\eta}_F\, (2\log b)^2 \;+\; O\Big(\tfrac{|\eta_F'|}{s\bar{\eta}_F} + \tfrac{|\eta_F''|}{s^2\bar{\eta}_F}\Big)\,.} \tag{43}$$

The full $\eta_F(\phi)$-local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F/\alpha)[\eta_F''(\phi) - 2s\eta_F'(\phi) + s^2\eta_F(\phi)]$.

Suppression is $\mathbf{L}^1$/CDF, not pointwise. The particular solution $\delta\rho_F^{\text{part}} = \gamma(b)\, q_s/(s^2 - \tau^2)$ combined with (43) gives pointwise amplitude

$$\delta\rho_F^{\text{part}}(\phi) \;\approx\; \frac{\lambda_F \bar{\eta}_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, e^{-s\phi}, \tag{44}$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta\rho_F^{\text{part}}$ is *not* pointwise small in s. The true suppression comes from the boundary-layer width of q_s: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Hence

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \;\leq\; C_{\text{bc}}^{(1)}(\tau)\, \frac{\lambda_F \bar{\eta}_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, \frac{1 - e^{-s}}{s} \;=\; O\Big(\tfrac{1}{\log b}\Big)\,, \tag{45}$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \;\leq\; C_{\text{bc}}^{(2)}(\tau)\, \frac{\lambda_F \bar{\eta}_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, \sqrt{\tfrac{1 - e^{-2s}}{2s}} \;=\; O\Big(\tfrac{1}{\sqrt{\log b}}\Big)\,, \tag{46}$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_\infty \leq \|\delta\rho_F\|_1 = O(1/\log b)$. Fisher forcing is therefore globally small in CDF / L^1 transport at large b, but not necessarily pointwise small near $\phi = 0$.

Warp amplification at large τ. Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_\infty \;\leq\; \frac{\|\delta\rho_F\|_1}{\inf_\phi \rho_{\text{full}}(\phi)}, \qquad \inf_\phi \rho_{\text{EVQ}}(\phi) \;=\; \frac{\tau}{\sinh\tau}, \tag{47}$$

so CDF perturbations are amplified by $\sinh\tau/\tau$ when translated into warp errors; very short L or very large d_{head} should re-include the forcing branch.

Dimensionless diagnostic. Using $\|\rho_{\text{EVQ}} - 1\|_2 = \tau^2/\sqrt{45} + O(\tau^4)$ and $\Gamma_b := (\lambda_F \bar{\eta}_F/\alpha)\, s^2/((s^2 - \tau^2)\tau^2)$, the L^2-level ratio is

$$R_F \;:=\; \frac{\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2}{\|\rho_{\text{EVQ}} - 1\|_2} \;\leq\; C_{\text{bc}}^{(2)}\, \Gamma_b\, \sqrt{45}\, \sqrt{\tfrac{1 - e^{-2s}}{2s}} \;=\; O\Big(\tfrac{1}{\sqrt{\log b}}\Big)\,. \tag{48}$$

Second-order utility deficit. Writing the surrogate objective as a quadratic form with operator A_τ, the full branch improves the objective by

$$\Delta_F \;=\; \tfrac{1}{2}\gamma(b)^2\,\langle q_s^\perp,\, A_\tau^{-1} q_s^\perp\rangle, \tag{49}$$

so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \le r$, the missed correction is $O(r^2)$ relative to the tether deformation energy.

Practical rule. Retain the forcing branch whenever the measured R_F exceeds a tolerance r_{tol} (we use 0.05 for training, 0.01 for theory plots), or when $\tau/(2\log b)$ is not small. The numerical threshold depends on the paper's Fisher normalization $\lambda_F\bar{\eta}_F/\alpha$, which we treat as a measurable constant rather than a derived value.

A.16 Channel-load motivation for Pearson χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head}-normalized stiffness $S_{\chi^2}(\tau){:=}\frac{1}{d_{\mathrm{head}}}\int_0^1(1{-}\rho_\tau)^2/\rho_\tau\,d\phi$, with leading expansion $\tau^4/(45\,d_{\mathrm{head}}){+}O(\tau^6)$ consistent with Proposition 1. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\mathrm{head}}\,S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha{=}1/d_{\mathrm{rot}}$ ($=1/d_{\mathrm{head}}$ for MHA via §A.10; $=1/d_{\mathrm{rope}}$ for compressed-RoPE MLA via §C.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi){=}1/\rho(\phi)$ and samples channels according to $dP_\rho{=}\rho\,d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) \;=\; \mathrm{Var}_{\phi\sim\rho\,d\phi}\Big[\tfrac{1}{\rho(\phi)}\Big] \;=\; \int_0^1 \frac{(1-\rho(\phi))^2}{\rho(\phi)}\,d\phi \;=\; S_{\chi^2}(\rho). \tag{50}$$

We treat this as a canonical motivated choice under the stated axioms, not a uniqueness claim.

Load-moment exponent. Distinct from the weighted-L^2 family $S_p(\tau){=}\frac{1}{d_{\mathrm{head}}}\int(\rho_\tau{-}1)^2/\rho_\tau^p\,d\phi$ used in Table 9 (where Pearson χ^2 is $p{=}1$), we also examine a load-moment family $S_p^{\mathrm{mom}}(\rho){=}\mathbb{E}_\rho[(1/\rho{-}1)^p]$. Substituting the EVQ cosh density and minimizing $S_p^{\mathrm{mom}}{-}\lambda U$ gives $\tau^{2(p-1)} \asymp d_{\mathrm{head}}^2/L_{\mathrm{eff}}^J$, hence $\tau^* \propto (L_{\mathrm{eff}}^J)^{-1/(2p-2)}$; only $p{=}2$ yields the observed $L^{-1/2}$ exponent within this family. The two families parametrize different f-divergence neighborhoods and need not match on the finite-τ grid.

***f*-divergences at finite τ.** Smooth f-divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho{=}1$ and share the leading $-1/2$ exponent asymptotically. Their *finite-τ* best points over the $L\in[128,4096]$ grid used in Table 9 fit exponents in $[0.58, 0.81]$, outside the $\pm1\%$ PPL basin around $\gamma{=}0.500$ at the tested scales.

A.17 λ as a curvature ratio

Invariant definition. Fix the normalizations of S_{χ^2} and U; then

$$\lambda_{\mathrm{true}}(\tau,L,b) \;=\; \frac{\partial_\theta S_{\chi^2}(\rho_{\sqrt{\theta}})}{\partial_\theta U_{\mathrm{exact}}(\rho_{\sqrt{\theta}};L,b)}\bigg|_{\theta=\tau^2}. \tag{51}$$

For the pure-tether branch, the un-normalized integral satisfies $d_{\mathrm{head}}\,S_{\chi^2}(\tau) = (\sinh\tau/\tau^2)\arctan(\sinh\tau) - 1 = \tau^4/45 + O(\tau^6)$ (cf. "Normalization convention" in §A.16). Writing $U_{\mathrm{exact}} = u_2^{\mathrm{exact}}(L,b)\,(d_{\mathrm{head}}^2/L_{\mathrm{eff}}^J)\tau^2 + O(\tau^4)$,

$$\lambda_\infty \;:=\; \lim_{\tau\to0}\lambda_{\mathrm{true}} \;=\; \frac{u_2^{\mathrm{exact}}}{u_2^{\mathrm{unit}}}. \tag{52}$$

A posteriori consistency check. Under the λ=1 unit convention, the surrogate prediction is $c_{\text{pred}}(L,b) = \sqrt{45\,Q_1(L,b)}$. Direct numerical integration of $Q_1(L,b) = \int_0^1 a(\phi)q(Lb^{-\phi})\,d\phi$ at b=500K, $L \in \{256, 512, 1024\}$ gives $Q_1 \in [0.0314, 0.0319]$ (near-constant on this grid), hence $c_{\text{pred}} \approx 1.19$. The exact-kernel collision minimizer c_{coll}=1.171 ± 0.001 (9 configurations, Table 7) agrees with this to within 2% (mean $c_{\text{coll}}/c_{\text{pred}}$=$0.981 \pm 0.003$, CV 0.28%), so $(c_{\text{coll}}/c_{\text{pred}})^2 = 0.96 \pm 0.005$ is within $\sim 4\%$ of unity. This is an *a posteriori* consistency check for the λ=1 convention inside the flat empirical basin; we do not treat λ as an independently derived physical constant.

Validation protocol (further). A tighter check would compute $u_2^{\text{exact}} = \partial_{\tau^2} U_{\text{exact}}(\rho_\tau; L, b)|_{\tau=0}$ directly from the oscillatory RoPE kernel and compare against $u_2^{\text{unit}} = Q_1(L,b)$; the collision-score proxy above suggests the ratio is already within 4% of unity across our configurations.

A.18 Discrete-channel transport gap and the MLA channel-scarcity mechanism

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi)$=$\int_0^\phi \rho$, $Q(u)$=$F^{-1}(u)$, midpoint grid u_k=$(k-\frac{1}{2})/K$, and ϕ_k=$Q(u_k)$. The empirical channel measure is μ_K=$K^{-1}\sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \le \rho \le M$ and $\|\rho'\|_\infty \le B$; then Q is $(1/m)$-Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \;\le\; \frac{1}{2Km}, \qquad W_1(\mu_K, \mu_\rho) \;\le\; \frac{1}{4Km}. \tag{53}$$

Defining the quantile-cell histogram $\rho_K(\phi)$=$1/(K|I_k|)$ on I_k=$[Q((k-1)/K),\, Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \;\le\; \frac{B}{Km}, \qquad \|\rho_K - \rho\|_\infty \;\le\; \frac{B}{Km}. \tag{54}$$

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$, $m_\tau = \tau/\sinh\tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \;\le\; \frac{\sinh\tau}{4K\tau}, \qquad \|\rho_{K,\tau} - \rho_\tau\|_1 \;\le\; \frac{\tau\sinh\tau}{K} \;=\; \frac{\tau^2}{K} + O\Big(\tfrac{\tau^4}{K}\Big). \tag{55}$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left|\iint K_{\text{sm}}\,d\mu_K\,d\mu_K - \iint K_{\text{sm}}\,d\mu_\rho\,d\mu_\rho\right| \;\le\; \frac{L_K}{2Km} + O(K^{-2}). \tag{56}$$

The singular δ-component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w,

$$\mathcal{D}_K[\rho] \;\approx\; \frac{1}{12K^2}\int_0^1 \frac{w(\phi)}{\rho(\phi)^2}\,d\phi + O(K^{-3}). \tag{57}$$

This is the finite-channel mechanism: a shaped ρ reduces the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. Going from standard MHA (K=64) to MLA (K=16, i.e. d_{rope}=32) scales the K^{-2} component of the distortion bound by a factor of 16 and the K^{-1} component by a factor of 4. The observed PPL-gain ratio between MLA (31.1% at K=16; §4.4) and MHA (−13.3% at K=64, 3-seed 454M FineWeb-Edu row in Table 21; with 1-seed rows ranging −10.9% to −18.9%) is approximately 2×: this is directionally consistent with the predicted K^{-1}/K^{-2} amplification but is not a quantitative match, since the realized PPL gain is moderated by architecture, optimization dynamics, and loss nonlinearities that the surrogate does not model. The bounds upgrade the Class C2 observation from a qualitative "channels-scarce regime amplifies shape correction" to an explicit K^{-1}/K^{-2} scaling on the surrogate distortion.

Caveat. At large τ (short L or large d_{head}), $m_\tau = \tau/\sinh\tau$ becomes small and the $1/(Km_\tau)$ factor grows, so these bounds become vacuous in the very-large-τ regime where the pure-tether branch itself needs re-inclusion of the forcing term (§A.15).

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 11: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	512	454M	3 + 3	FineWeb-Edu + 10% passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Table 12 lists the training hyperparameters shared across text experiments; per-tier deviations (learning rate, batch size) are noted.

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500$K, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.1 to 24.4 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 454M comparisons used in the main text.

Table 12: Training hyperparameters for the primary text experiments. All tiers share the same optimizer and schedule; per-tier values are noted where they differ.

Parameter	Value
Optimizer	AdamW (β_1=0.9, β_2=0.95, ϵ=10^{-8})
Learning rate	6×10^{-4} (50M/125M), 2×10^{-4} (454M), 1.5×10^{-4} (750M)
LR schedule	Cosine decay to 0.1× peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0 (no dropout)
Sequence packing	Contiguous, no cross-document masking
Video DiT (129M–382M)	
Optimizer	AdamW (β_1=0.9, β_2=0.95)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
MLA (432M)	
LR / warmup	6×10^{-4} / 100 steps
Training tokens	500M

Table 13: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

B.4 Progressive training dynamics

Table 14 and Figure 5 detail the supporting progressive training result referenced in the body. This is a single-seed run separate from the primary EVQ×YaRN anchor above, whose training length is 512. The key observation is the divergence pattern: EVQ+YaRN PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). In this single-seed setup the gap is consistent with being driven by geometric RoPE's progressive degradation rather than by further intrinsic EVQ improvement.

Table 14: Progressive training widens EVQ's advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	−34.6%
2	512→1024	4.60	2.21	−52.0%
3	512→1024→2048	13.17	2.48	−81.2%
Stage 3: PPL@48K		14.22	**2.63**	−82%

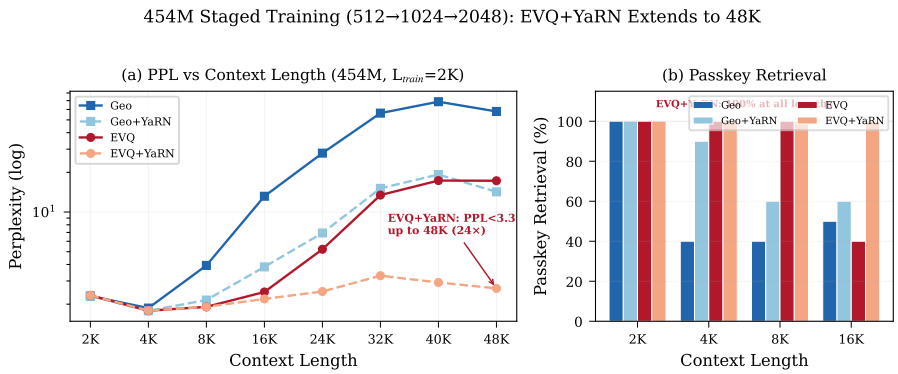

Figure 5: **Progressive training widens EVQ's advantage.** Three-stage context extension ($512 \to 1024 \to 2048$, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K (24× training length) at PPL 2.63.

B.5 Modality-dependent τ correction for video DiT

The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For bidirectional DiT, the best observed sweep point for τ is approximately 0.53× the text prediction (τ=1.5 vs. predicted 2.83 for K_t=16, T=32). A schedule-aware decomposition writes the ratio as

$$\frac{\tau_{\text{video}}}{\tau_{\text{causal}}} \approx \frac{1}{\sqrt{2}} \sqrt{\mathbb{E}_t[\alpha_t^m]}, \tag{58}$$

where $1/\sqrt{2}$ is the bidirectional softmax-Jacobian factor (a diffuse bidirectional window of size $2L$+1 halves the directional curvature relative to a causal window of size L; Appendix A.14), and $\sqrt{\mathbb{E}_t[\alpha_t^m]}$ is a schedule-dependent signal-amplitude contraction with m determined by the utility definition. Our training run uses a Rectified-Flow (RF) schedule α_t=1−t with uniform $t \sim \mathcal{U}[0,1]$ and no loss weighting (not a VP SDE). Under RF, $\mathbb{E}_t[\alpha_t]$=0.5, $\mathbb{E}_t[\alpha_t^2]$=1/3, and $\mathbb{E}_t[\alpha_t^4]$=0.2, giving predicted ratios 0.500, 0.408, and 0.316 for $m \in \{1, 2, 4\}$ respectively. Only m=1 (linear signal-expectation utility) matches the empirical 0.53 within 6%; the covariance-linear m=2 is 23% below observation. We therefore report (58) as a directional decomposition and leave a principled utility-exponent derivation under RF (specifying why m=1 rather than m=2 is the correct contraction for the RoPE phase-coherence utility) as future work.

B.6 Cross-modal evidence: video DiT with bidirectional attention

To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST [Srivastava et al., 2015], training on 32 frames and extrapolating to 128 frames (4×) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and CUDA non-determinism—controlling for shared optimizer state and CUDA nondeterminism and isolating the implemented frequency-allocation difference within this setup.

Table 15 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by −26%, all-frame extrapolation by −15%, and far-frame extrapolation by −32%. The consistency across both seeds—EVQ wins all three metrics in each—suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds) is too small for formal statistical testing.

This supporting result probes EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design eliminates all confounders except the frequency allocation

Table 15: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	−21%
42	All extrap	0.00670	**0.00573**	−14%
42	Far extrap	0.00774	**0.00502**	−35%
137	Train MSE	0.01113	**0.00761**	−32%
137	All extrap	0.00713	**0.00598**	−16%
137	Far extrap	0.00856	**0.00601**	−30%
Mean Δ				−26% / −15% / −32%

itself. A preliminary scale-up to 382M DiT (3× larger, single seed) yields directionally consistent results (−35% far-frame MSE), and EVQ outperforms Geo under all three inference-time methods tested (YaRN, RIFLEx, raw). This is reported as supporting evidence only; multi-seed replication is needed to confirm the pattern at this scale.

Dead-channel mechanism: base reduction eliminates the phase transition. At base = 10,000 with K_t=16 temporal frequencies and 32 training frames, the lowest-frequency channels accumulate less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead. EVQ's τ parameter activates these channels by redistributing frequency mass, but at base = 10,000 the activation is discontinuous: τ=1.2 leaves 11/16 channels alive (effective rank 9), while τ=1.5 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 15.

To test this mechanism, we repeat the head-to-head experiment with base = 1,000 (same architecture, same run; 2 seeds). At this base, the geometric allocation already starts with 13/16 alive channels (rank 10), so the dead-channel bottleneck is resolved. Table 16 is consistent with the prediction: (1) τ=1.2 fully recovers, winning Geo by −48% far-frame MSE (vs. +162% at base = 10,000); (2) τ=1.2 and τ=1.5 are statistically indistinguishable ($<$ 0.3% difference on all metrics), supporting the interpretation that the phase transition was driven by the dead-channel threshold rather than by an intrinsic property of τ=1.2.

Table 16: Dead-channel mechanism check: base = 1,000 DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between τ=1.2 and τ=1.5 observed at base = 10,000 is no longer present in this seed; both EVQ configurations win Geo by ≈48% far-frame MSE.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	**0.00933**	**0.00820**	**0.00859**	−48%
1.5	**0.00930**	0.00822	**0.00858**	−48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 17 audits five major video DiT architectures, all of which inherit base = 10,000 from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k-th temporal frequency. Channels with phase $<$ 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315× shorter than text sequences (≥4096 tokens) for which base = 10,000 was designed. The text reference (LLaMA-2) has zero dead channels, consistent with the dead-channel problem being specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 16 shows that it remains effective regardless of the base choice.

Table 17: Dead temporal channels in major video DiT architectures. All models inherit base = 10,000 from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B [Yang et al., 2024b]	8	13	4/8	50%	56×
Wan-2.1 1.3B [Wan Team, 2025]	22	21	9/22	41%	42×
Latte-1 [Ma et al., 2024]	12	16	5/12	42%	50×
Open-Sora 1.2 [Zheng et al., 2024b]	22	51	7/22	32%	17×
HunyuanVideo [Kong et al., 2024]	8	33	3/8	38%	21×
Our DiT	16	32	6/16	38%	26×
LLaMA-2 7B [Touvron et al., 2023] (text)	64	4096	0/64	0%	—

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-channel severity. Both GEO (τ=0) and EVQ (τ=1.5) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4× temporal extrapolation.

Table 18: Base sweep: far-frame denoising MSE (t=0.5, 4× extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across base≥500; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at base = 5,000.

	With YaRN			Without YaRN			
Base	GEO	EVQ	Δ	GEO	EVQ	Δ	Dead ch.
100	**0.0161**	0.0189	+18%	**0.296**	0.356	+20%	0/16
500	0.0210	**0.0207**	−1%	0.224	**0.146**	−35%	∼3/16
1000	0.0517	**0.0178**	−66%	0.286	**0.132**	−54%	∼5/16
5000	0.0787	**0.0210**	−73%	0.172	**0.111**	−36%	∼5/16
10000	0.0401	**0.0137**	−66%	0.107	**0.048**	−55%	6/16
50000	0.1895	**0.0262**	−86%	0.247	**0.108**	−56%	∼9/16

Two key findings emerge in this 1–2 seed video-DiT setting. First, when all channels are alive (base = 100, 0/16 dead), GEO wins by ≈20%, consistent with EVQ's advantage being specific to the dead-channel regime. Second, EVQ+YaRN is stable across base≥500 (1.9× MSE range vs. GEO's 12×), indicating robustness to the base hyperparameter within the tested grid. The best observed grid point is EVQ at base = 10,000 with YaRN (0.0137), better than GEO at any tested base. A pure training-length evaluation (32f, no extrapolation) shows all 12 configurations produce near-identical MSE (< 6% variation), supporting the interpretation that the advantage is an extrapolation phenomenon rather than a training-distribution effect.

C Supporting Results

C.1 Multi-head Latent Attention

To test whether EVQ's benefit extends to Multi-head Latent Attention (MLA), the compressed-RoPE attention variant introduced in DeepSeek-V2 [DeepSeek-AI, 2024a], we train a 432M-parameter MLA Transformer from scratch. Our configuration uses d_{rope}=32 and base = 500K; production DeepSeek-V2/V3 [DeepSeek-AI, 2024b] uses d_{rope}=64 and base = 10K. The shared architectural property—a compressed RoPE subspace with far fewer frequency channels than standard MHA—is what makes allocation quality decisive. Only 16 frequency channels carry positional information, while the remaining kv_rank=256 dimensions carry content. (d_{rope} in MLA notation is the same quantity as d_{rot} in §3; we use the DeepSeek-style d_{rope} throughout this subsection.)

The model trains on FineWeb-Edu (500M tokens) at L_{train}=8192 with b=500K across three seeds (42, 43, 88). EVQ uses τ=1.414 (i.e., $d_{\text{head}}/\sqrt{L} = 128/\sqrt{8192}$).

Two distinct dimensions. It is important to distinguish two roles played by the head dimension. (i) The *kernel normalization* $\alpha=1/(2K)=1/d_{\text{rope}}$ controls the diagonal coefficient of K_{app} and depends only on the number of RoPE frequency pairs ($K=d_{\text{rope}}/2$); for MLA this gives $\alpha=1/32$. (ii) The *operating-point dimension* d_{eff} in $\tau^*=d_{\text{eff}}/\sqrt{L}$ enters through the post-softmax transport utility U, which weights all attention-logit components carried by the head—including the non-RoPE content channels mixed in via the latent projection. The variational $\alpha=1/d_{\text{rope}}$ derivation alone would give a much smaller deployment τ; we deploy $d_{\text{eff}}=d_{\text{head}}=128$ as a calibrated convention reflecting the latent-projection coupling, validated empirically by Table 19 but not derived from the broadband surrogate. A direct $\tau=d_{\text{rope}}/\sqrt{L}$ ablation is the natural sanity check.

Table 19: MLA validation (432M, d_{rope}=32, 3-seed mean ± std). EVQ alone outperforms GEO+YaRN(s=4) at 16K and ties within printed precision at 24K (both 204.9). EVQ+YaRN is the best at every tested length.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4±0.9	138.8±5.5	241.5±2.6	323.7±5.2
GEO+YaRN(s=4)	35.5±0.9	117.9±6.5	204.9±6.9	278.5±6.3
EVQ	35.8±0.8	**95.6±4.1**	204.9±14.7	291.6±20.5
EVQ+YaRN(s=4)	35.8±0.8	**71.1±4.1**	**153.2±12.5**	**236.6±15.9**

Table 19 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% (138.8 → 95.6) with only +1.1% in-distribution cost—a larger relative gain than observed with standard MHA at comparable scales, consistent with the "fewer channels, each more precious" hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (−31.1% vs −15.1%); at 24K the two are tied within printed precision (204.9 each). Under matched YaRN scale in this setup, a training-time allocation change is thus comparable to or better than the inference-time rescaling baseline we tested, but we do not claim superiority over every possible inference-time method. Third, EVQ+YaRN is the best at every tested length; the ranking EVQ+YaRN > EVQ > GEO+YaRN > GEO holds strictly at 16K, with YaRN providing larger marginal benefit on EVQ (−25.6%) than on GEO (−15.1%) at 16K, consistent with complementary mechanisms.

Training-progression analysis shows EVQ's advantage is visible from early training and grows monotonically to −31.1% at completion, while the in-distribution cost simultaneously decreases from +1.4% to +1.1%. This pattern is consistent with the benefit being an intrinsic property of the frequency allocation rather than a training-dynamics artifact.

C.2 Matched-scale YaRN leverage (supporting table)

C.3 Multi-scale raw length generalization

C.4 Attention distance redistribution

Figure 6 illustrates the waterbed crossover in learned attention patterns of the 750M continue@4K checkpoint, complementing the surrogate-level analysis in §A.7.

C.5 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints (n=2086, distractor-padded, random article position 20%–80%). Table 22 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, consistent with 454M capacity—not positional encoding—being the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants: EVQ's NLL advantage grows from −1.7% at 4K (in-distribution) to −30.1% at 8K (2× extrapolation)

Table 20: Matched-scale YaRN leverage at $L = 256$; NTK-aware scaling is not complementary at large τ.

Method	YaRN gain @4K	YaRN gain @8K
Geo	$-4.5\% \pm 1.2$	$-3.1\% \pm 1.6$
EVQ $\tau = 2.0$	$-27.0\% \pm 0.1$	$-28.9\% \pm 1.3$
EVQ $\tau = 4.0$	$\mathbf{-32.5\% \pm 2.0}$	$\mathbf{-40.7\% \pm 1.6}$
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

Table 21: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
454M	TinyStories (Hybrid)	1	-0.4%	-13.7%
454M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+1.2%	-45.9%

and -21.4% at 16K (4×). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

Table 22: QuALITY QA downstream evaluation (454M, n=2086, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	$\mathbf{-30.1\%}$
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

Figure 8 visualizes the NLL divergence across context lengths, showing that EVQ's advantage grows with extrapolation ratio.

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body: EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

C.6 Multi-scale waterbed verification

Figure 9 visualizes the waterbed effect across model scales: EVQ trades a small in-range cost (±1.7%) for long-range gains (10–46%) across the tested grid.

D Supporting Experiments (from Main Text)

The following results extend EVQ beyond the primary MHA setting. Each is scoped to a specific question and labeled with its seed count.

MLA: allocation matters more when channels are scarce. EVQ extends to MLA [DeepSeek-AI, 2024a,b], where only 16 frequency channels carry positional information. On a 432M model (3-seed, 500M tokens, L_{train}=8192), EVQ alone reduces 2× extrapolation PPL by 31.1% and outperforms Geo+YaRN at 16K. Preliminary evidence under thorough training (1B tokens, single seed) suggests that EVQ's raw advantage reverses (+11.1%) while EVQ+YaRN after fine-tuning still wins (-2.5%)—a 13.6pp swing consistent with the

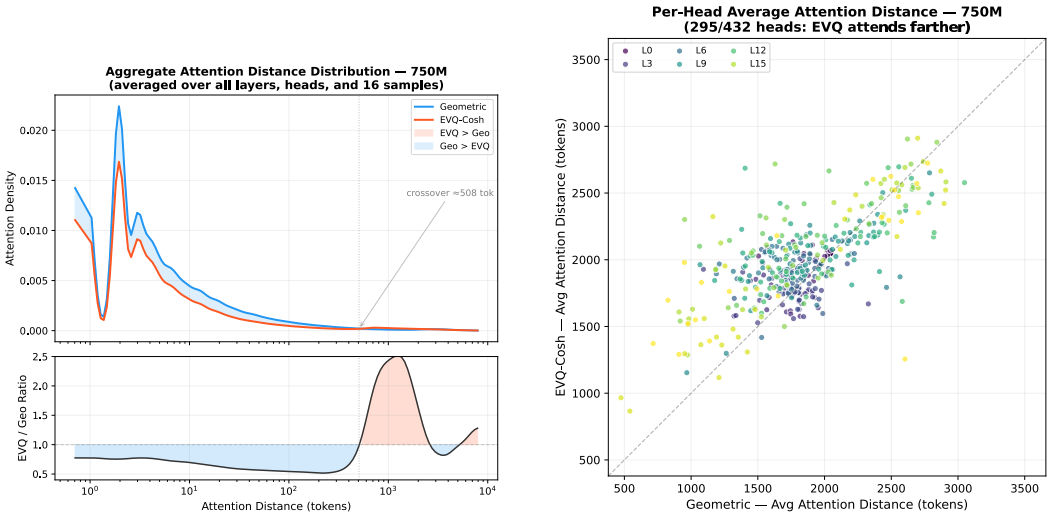

Figure 6: **Waterbed illustration in learned attention patterns (750M, single seed).** Left: aggregate attention density vs. token distance. A crossover at ≈508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to 2.5× higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

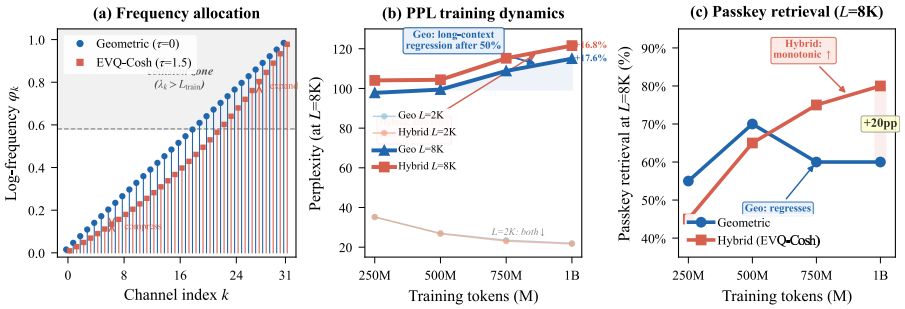

Figure 7: **750M frequency dynamics and retrieval divergence** (single seed). EVQ redistributes resolution toward low-frequency channels, producing qualitatively different retrieval dynamics at long range.

frequency substrate remaining the binding constraint even when raw extrapolation saturates. Multi-seed replication is in progress. Full 3-seed standalone results are in Appendix C.1.

Post-training injection via LoRA (7B scale). As supporting post-hoc evidence, we inject EVQ-Cosh frequencies (τ=1.414, the τ^* predicted for d_{head}=128, L=8192) into LLaMA-3-8B-Instruct [Grattafiori et al., 2024] and fine-tune with LoRA (r=64=K, α=128) on LongAlign-10k for 300 steps.

Table 24 shows that EVQ-LoRA reduces extrapolation PPL by 8× at 16K and 19× at 32K with a modest in-distribution cost (+30% PPL at 8K), serving as supporting post-hoc evidence at 7B scale. The rank is set at r=$d_{\text{head}}/2$=64. The phenomenological variational model in §A.12 predicts a rank-dependent viability threshold near $r \approx d_{\text{head}}/2$ under our calibration: below the threshold, coupling stiffness appears to prevent frequency redistribution; above it, extrapolation recovers. RULER scores do not improve, consistent with LoRA capacity sufficing for positional adaptation but not instruction-following at extrapolated lengths; a matched Geo+LoRA+LongAlign control on identical training setup is the natural follow-up to fully isolate the EVQ effect from LoRA capacity.

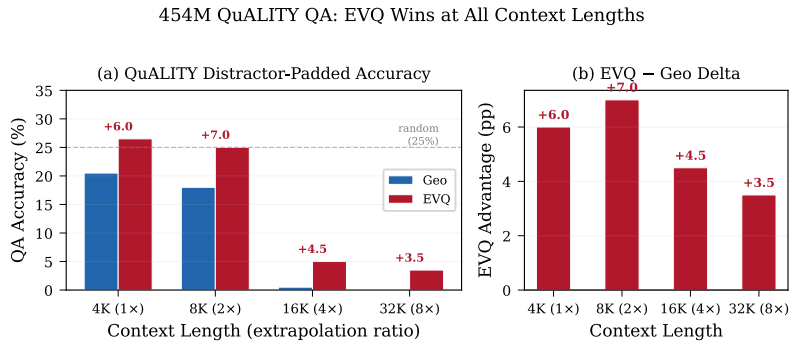

Figure 8: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, n=2086). EVQ's NLL advantage grows from −1.7% at in-distribution (4K) to −30.1% at 2× extrapolation (8K).

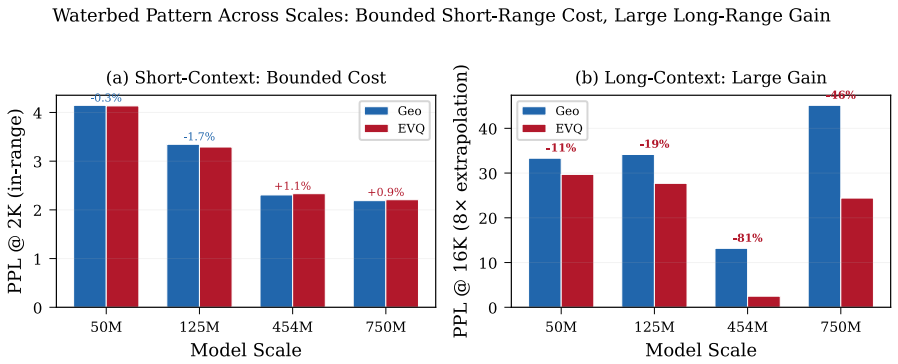

Figure 9: **Waterbed illustration across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving substantial long-range gains. The pattern is consistent from 50M to 750M within the tested grid, consistent with the theoretical prediction.

Progressive training (single seed, preliminary). Three-stage progressive training (454M, 512→1024→2048) is reported as supporting evidence. The EVQ+YaRN PPL gap at 16K widens from −34.6% to −81.2%, driven by Geo+YaRN degradation while EVQ+YaRN remains stable. At Stage 3, functional context reaches 48K (24× training length) at PPL 2.63. After Stage 2, EVQ *raw* (PPL 11.2 at 16K) surpasses EVQ+YaRN at Stage 1 (16.8), suggesting that continued training can partially internalize the allocation benefit. This is a single-seed observation; multi-seed replication is in progress.

Cross-modal mechanism check: video DiT. In 129.6M DiT head-to-head experiments (2 seeds, shared training run), EVQ wins every metric in both seeds, eliminating optimizer and CUDA confounds. Once the dead-channel bottleneck is removed (base-1000), both τ=1.2 and τ=1.5 beat Geo by ≈48% on far-frame MSE. This is a supporting mechanism check rather than a standalone video contribution: the collision bottleneck identified by the theory appears across modalities. Full results are in §B.5.

Signal gradient across abstraction layers. Across the diagnostics reported here, EVQ's effect generally attenuates through successive abstraction layers: raw PPL −46% (Table 21) → Gold-Answer NLL −30% (Table 22) → passkey +59pp vs. Geo baseline (EVQ+YaRN 100% vs. Geo 41%, 3 seeds) → QA accuracy +2.2pp (Table 22, single seed, near-random baseline). This pattern is consistent with the intervention being largest at the PE layer and attenuated through higher abstraction where model capacity becomes the binding constraint.

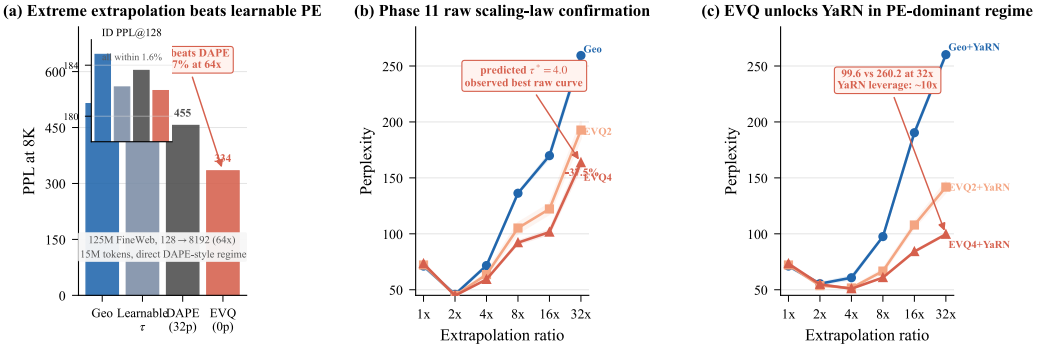

Figure 10: PE-dominant evidence (visualization of Table 5). Left: EVQ's closed-form allocation (zero learned parameters) closes more of the 128→8K extrapolation gap than the learned positional-operator baseline. Right: at L=256, the predicted τ^*=4.0 is favored among tested settings, and EVQ4+YaRN outperforms Geo+YaRN across extrapolation ratios.

Table 23: L=256 YaRN composition study (125M, PPL). The predicted τ^*=4.0 is the best of the tested settings, and EVQ4+YaRN outperforms Geo+YaRN at every extrapolation ratio. This supplements Table 5 in the main text.

Method (L=256)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

Capability preservation. Table 25 shows that EVQ is non-destructive when retrieval is absent from training and improves extrapolated retrieval by +10–13pp when retrieval is learned (454M, 3-seed).

E Broader Impact Statement

This work studies frequency allocation as a design axis for rotary position embedding rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing the default geometric frequency schedule with a principled closed-form allocation can reduce wasted compute and improve the reliability of long-context models across architectures. The main risk is over-generalization: our strongest evidence is concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

Table 24: EVQ-Cosh LoRA on LLaMA-3-8B-Instruct (r=64, 300 steps, single seed). EVQ trades a modest in-distribution cost for substantial extrapolation gains.

Context	Base PPL	EVQ-LoRA PPL	Δ
8K (in-dist.)	**7.42**	9.63	+30%
16K (2×)	176.3	**21.5**	−88%
32K (4×)	1942.5	**104.3**	−95%

Table 25: Capability preservation and retrieval robustness (454M, 3-seed, *per-document* PPL scoring; cf. Table 4 for full-sequence scoring on the same model). EVQ is non-destructive when retrieval is absent, improves extrapolated retrieval, and is more robust to denser passkey supervision. The absolute PPL@16K differs from Table 4 (253.2 vs. 262.0) because of the scoring protocol only; the relative Geo→EVQ ranking is unchanged.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% → 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

NeurIPS Paper Checklist

1. **Claims**
   - **Question:** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   - **Answer:** [Yes]
   - **Justification:** The abstract and introduction are restricted to three contributions: (1) identifying training-time frequency allocation as a PE design axis with a closed-form variational family, (2) demonstrating that a semi-analytic rule—with derived d_{head} and $L^{-1/2}$ structure and an empirically calibrated $\mathcal{O}(1)$ scale—falls within the best observed sweep basin across the tested grid of 27 configurations (primary claims rely on 3-seed anchors; supporting 1–2-seed rows extend scope), and (3) the matched-scale systems result that EVQ increases YaRN leverage under a fixed YaRN scale (we do not claim dominance over every tuned YaRN baseline). The τ^* rule's epistemic status—derived d_{head} and $L^{-1/2}$ structure under the stated small-τ scalings, channel-load-motivated χ^2 stiffness, and unit-convention λ—is stated explicitly in §3.7.
2. **Limitations**
   - **Question:** Does the paper discuss the limitations of the work performed by the authors?
   - **Answer:** [Yes]
   - **Justification:** The body explicitly marks the broadband surrogate as the main approximation, presents τ^* as a semi-analytic operating-point rule rather than a tuned hyperparameter, and demotes single-seed and video evidence to supporting status.
3. **Theory assumptions and proofs**
   - **Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   - **Answer:** [Yes]
   - **Justification:** Theorems 1–2 and Proposition 1 are exact conditional on the broadband surrogate; the surrogate itself is supported by a functional check (24–92% collision-score reduction across 12 configurations) in Appendix A.5. The τ^* scaling law (Proposition 1) is semi-analytic: d_{head} dependence and the $L^{-1/2}$ exponent are derived, but one transport constant λ is calibrated empirically. This status is stated explicitly in §3.7 and the limitations section.
4. **Experimental result reproducibility**
   - **Question:** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
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   - **Justification:** The body states the principal settings and the appendix records datasets, seeds, evaluation grouping, and the status of larger supporting runs.
5. **Open access to data and code**
   - **Question:** Does the paper provide open access to the data and code?
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   - **Justification:** We include an anonymous supplementary code archive containing the EVQ-Cosh inverse-frequency initializer, the collision-surrogate and τ^*-rule analyses, evaluation scripts, and configuration files for the primary experiments (EVQ×YaRN 454M, PE-dominant 125M/454M, MLA 432M). Upstream datasets (FineWeb-Edu, TinyStories, QuALITY, RULER passkey/NIAH) are available from their respective public releases; our passkey-mix composition and data-prep scripts are in the archive. Model checkpoints and weights are not included to preserve double-blind review and to keep the archive within size limits. A reproducibility README lists the commands to regenerate each primary-tier table and figure from a fresh checkout.

6. **Experimental setting/details**
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